

Evolution of Sonar and Sonar Signal Processing Fundamentals

Spencer J. Chang, PhD
Daniel D. Sternlicht, PhD

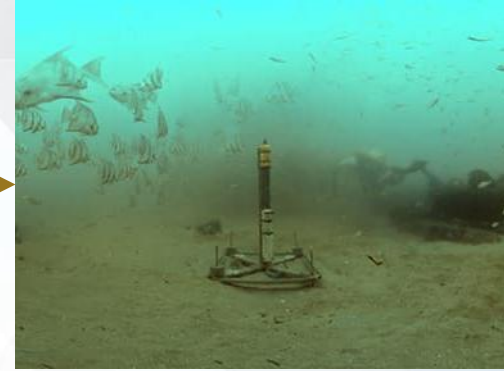
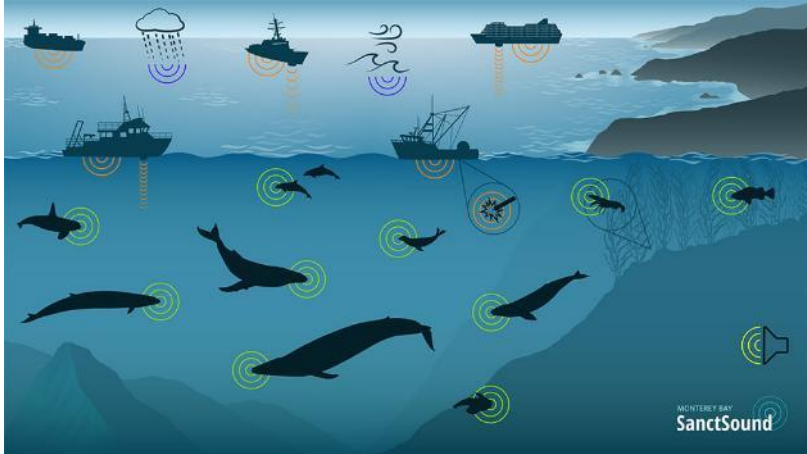
September 21, 2026

Distribution A. Approved for public release. Distribution is unlimited.

Content is largely based on standard reference materials.

Signal Characteristics

Signals – Sampling Rate



Analog Signal Processing
LC-type filters, Op-amps.

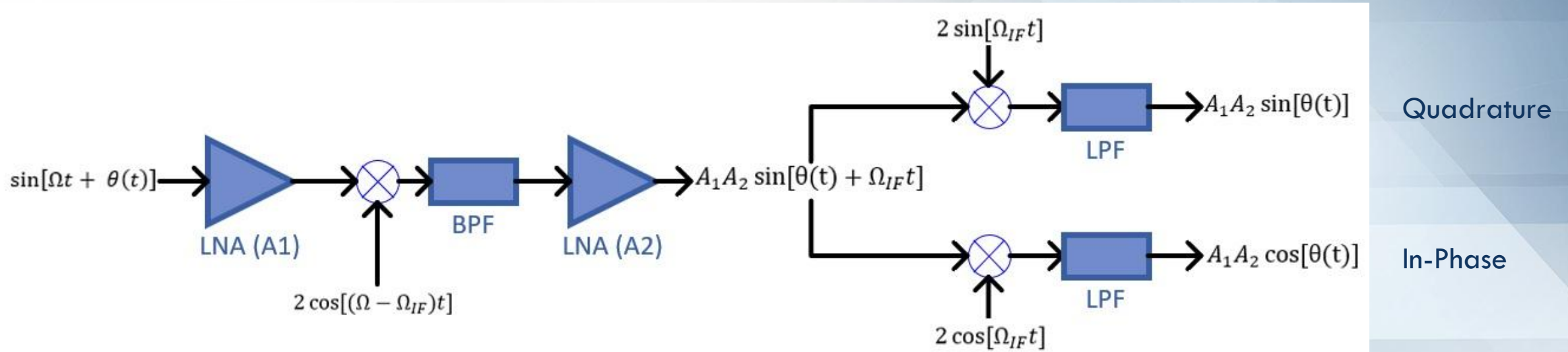
A/D Converter
(Digitize Signals)

Digital Signal Processing
w/Software

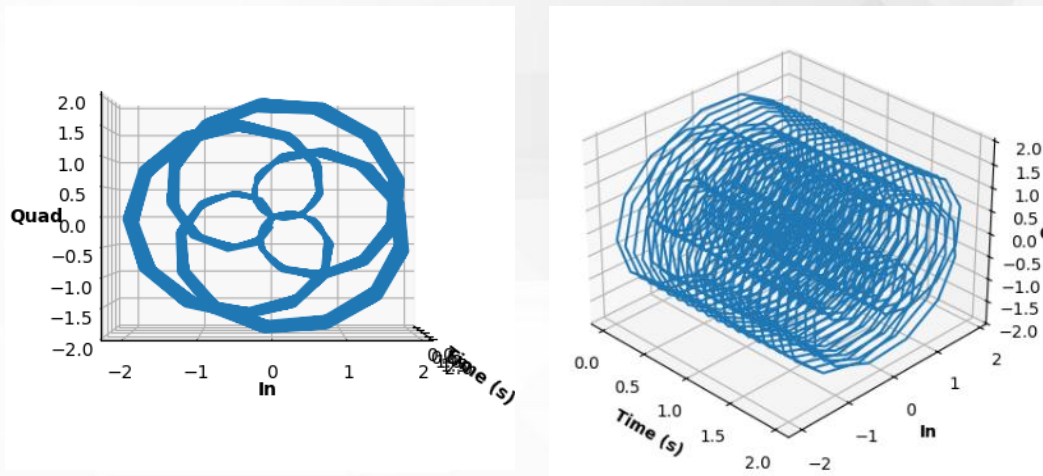
Left Image: <https://sanctuaries.noaa.gov/news/apr22/new-sanctsound-portal.html>

Right Image: <https://oceanservice.noaa.gov/facts/hydrophone.html>

Signals – Receiver Circuit

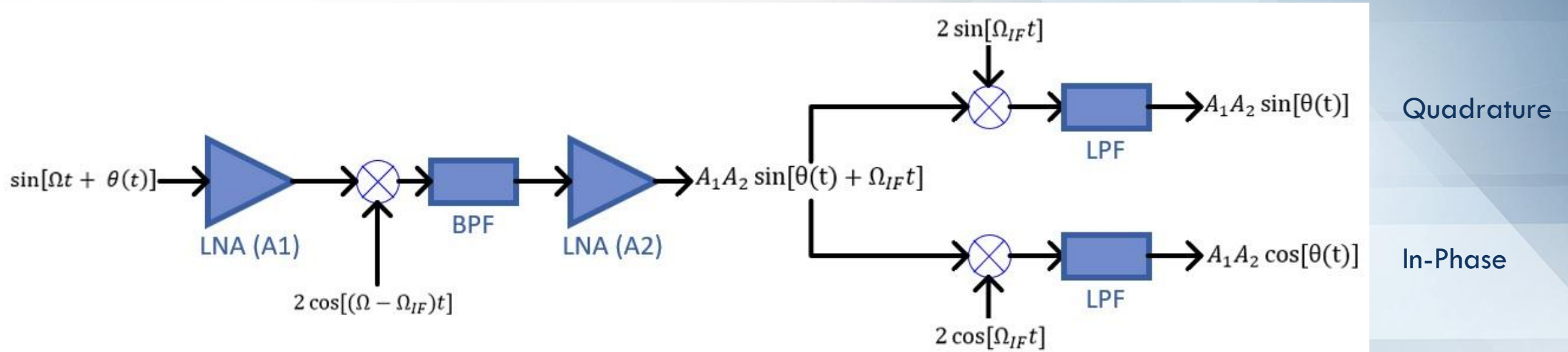


M. A. Richards, *Fundamentals of Radar Signal Processing*, 2nd ed., New York, NY, USA: McGraw-Hill Education, 2014.



$$A_1 A_2 (\cos[\theta(t)] + j \sin[\theta(t)]) = A_1 A_2 e^{j\theta(t)}$$

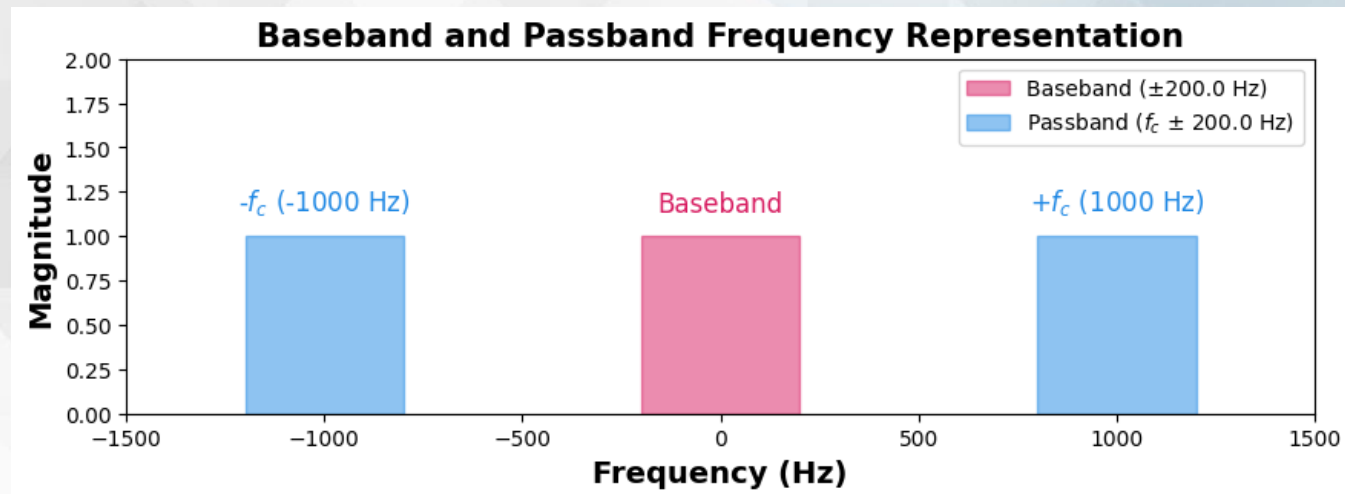
Signals – Passband vs. Baseband



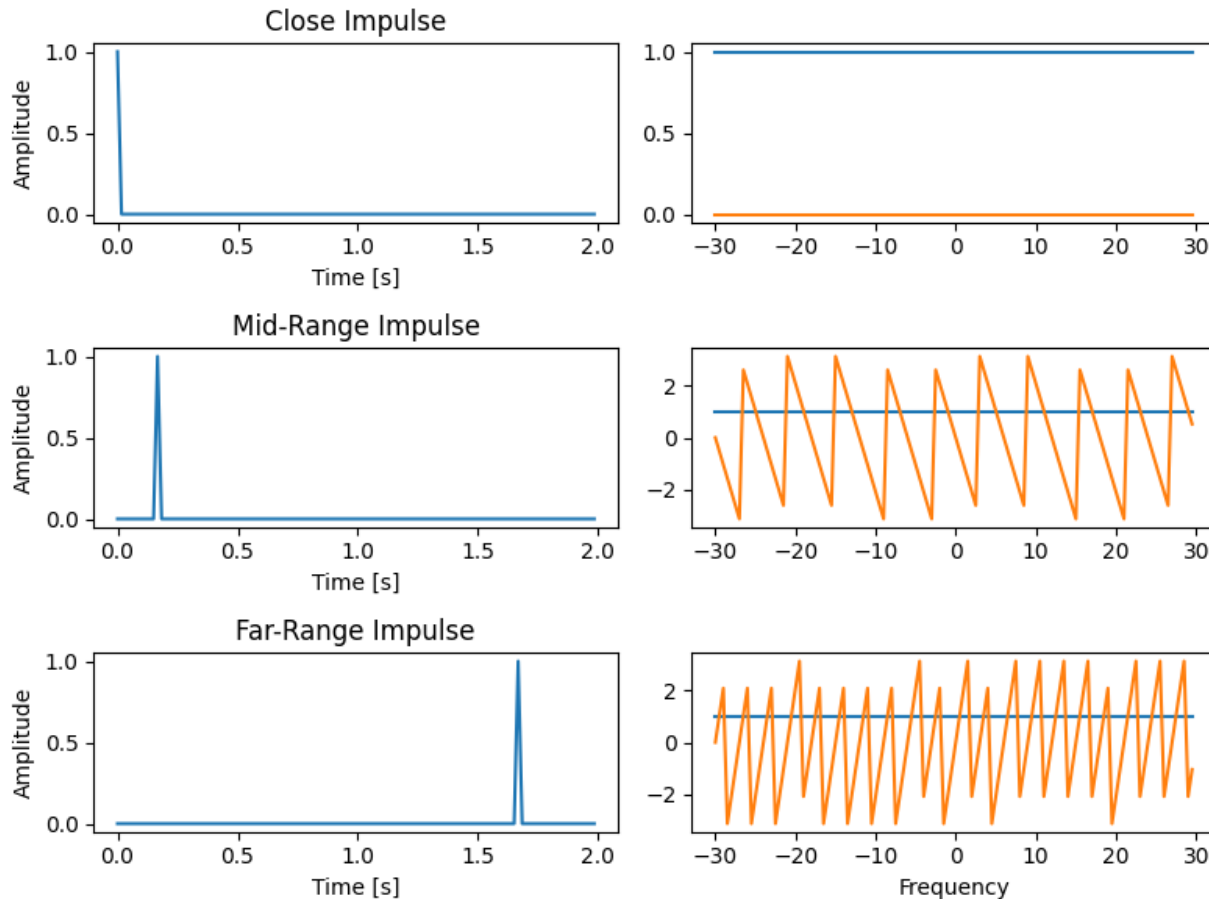
M. A. Richards, *Fundamentals of Radar Signal Processing*, 2nd ed. New York, NY, USA: McGraw-Hill Education, 2014.

$$\Omega = 2\pi f$$

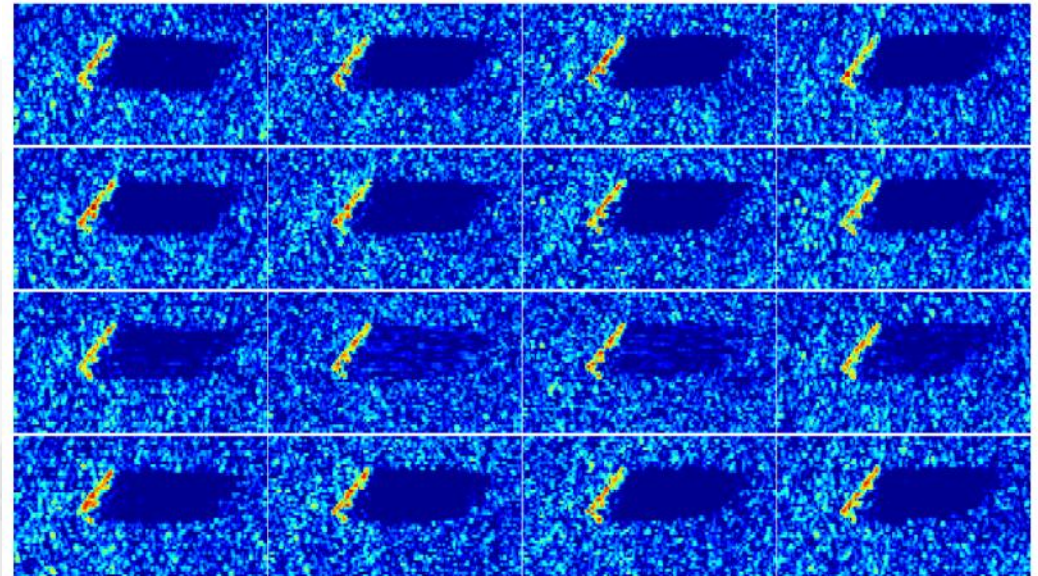
$$\Omega_{IF} = 2\pi f_{IF}$$



Signals – Phase Information



- Time-of-arrival
- Direction-of-arrival
- Classification of materials/seabeds
- Doppler shifting
- Interferometry



D. P. Williams and A. J. Hunter, "Multi-look processing of high-resolution SAS data for improved target detection performance," in Proc. IEEE Int. Conf. Image Process. (ICIP), Quebec City, QC, Canada, Sep. 2015, pp. 153–157.

Fourier Analysis

Fourier Analysis – Fourier Transform

Fourier Transform

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

$$\omega = 2\pi f$$

Fourier Series Synthesis

$$x(t) \cdot \phi_m(t) = \frac{1}{T} \int_0^T \left[\sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t} \right] \phi_m^*(t) dt$$

$$= \frac{1}{T} \int_0^T (a_m e^{jm\omega_0 t}) e^{-jm\omega_0 t} dt = a_m$$

Discrete Fourier Transform (DFT)

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{nk}{K}}, k \in \{0, 1, \dots, K-1\}$$

Inverse DFT

$$x[n] = \sum_{k=0}^{K-1} X[k] e^{+j2\pi \frac{nk}{K}}, n \in \{0, 1, \dots, N-1\}$$

Fourier Analysis – Intuition

Recall the concept of orthonormal basis vectors

$$\mathbf{u}_1 = [1, 0]^T$$

$$\mathbf{u}_2 = [0, 1]^T$$

$$\therefore [4.5, 0.2]^T = 4.5\mathbf{u}_1 + 0.2\mathbf{u}_2,$$

$$\text{where } \|[4.5, 0.2]^T\| = \sqrt{[4.5, 0.2] \cdot [4.5, 0.2]^T} \approx 4.504$$

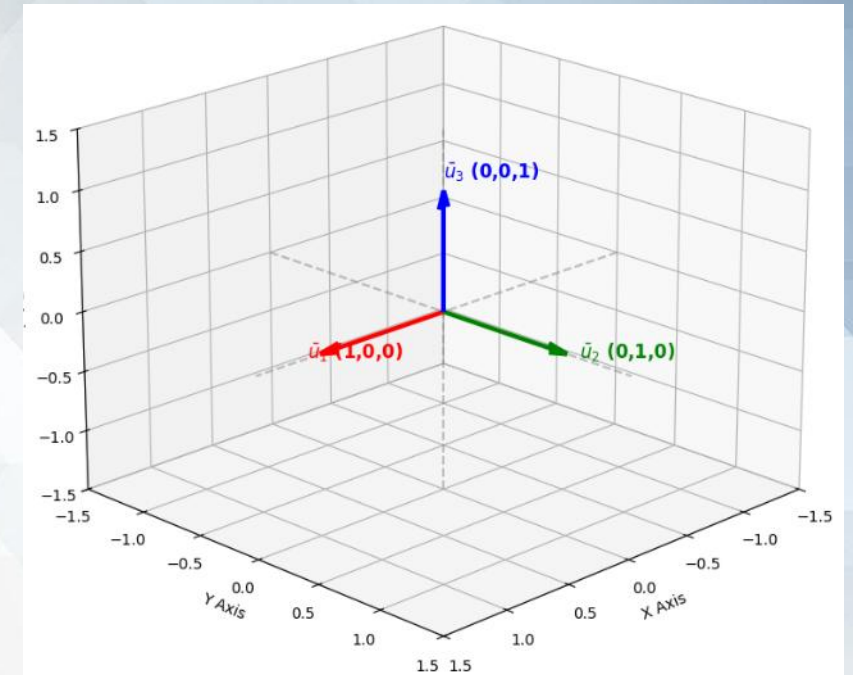
For a linear vector space of dimension 2 (2D),...

$$\mathbf{u}_1^T \mathbf{u}_1 = 1$$

$$\mathbf{u}_2^T \mathbf{u}_2 = 1$$

$$\mathbf{u}_1^T \mathbf{u}_2 = \mathbf{u}_2^T \mathbf{u}_1 = 0$$

This holds for any linear vector space of dimension n (i.e., it has n orthonormal basis vectors $\mathbf{u}_k, k \in \{1, \dots, n\}$).



Fourier Analysis – Intuition

Harmonic complex exponentials $e^{jn\omega_0 t}$ have similar properties when we define the inner product operation as an integral over the period T . Let us define each of these exponentials as $\phi_n(t) = e^{jn\omega_0 t}$.

Thus, given a $\phi_n(t)$ and $\phi_m(t)$,

$$\begin{aligned}\phi_n(t) \cdot \phi_m(t) &= \frac{1}{T} \int_0^T \phi_n(t) \phi_m^*(t) dt \\ &= \frac{1}{T} \int_0^T e^{jn\omega_0 t} e^{-jm\omega_0 t} dt \\ &= \frac{1}{T} \int_0^T e^{j(n-m)\omega_0 t} dt \\ \Rightarrow \phi_n(t) \cdot \phi_m(t) &= \begin{cases} 0, & \text{if } n \neq m \\ 1, & \text{if } n = m \end{cases}\end{aligned}$$

Fourier Analysis – Fourier Transform

Fourier Transform

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

$$\omega = 2\pi f$$

Fourier Series Synthesis

$$x(t) \cdot \phi_m(t) = \frac{1}{T} \int_0^T \left[\sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t} \right] \phi_m^*(t) dt$$

$$= \frac{1}{T} \int_0^T (a_m e^{jm\omega_0 t}) e^{-jm\omega_0 t} dt = a_m$$

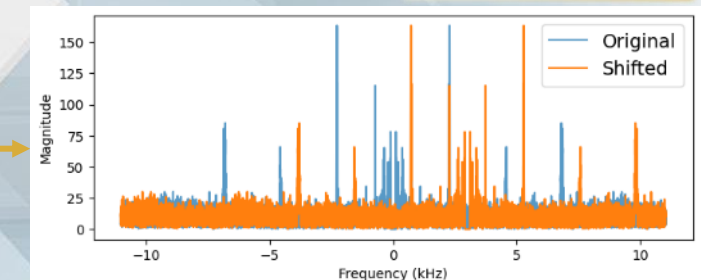
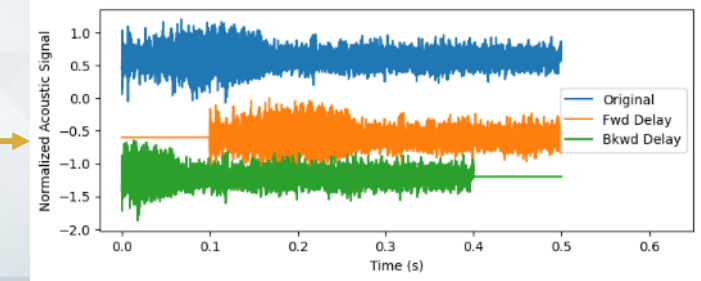
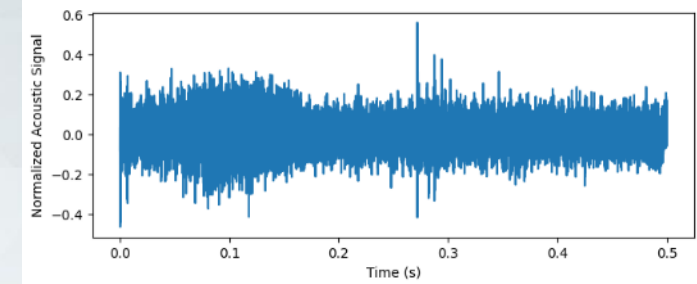
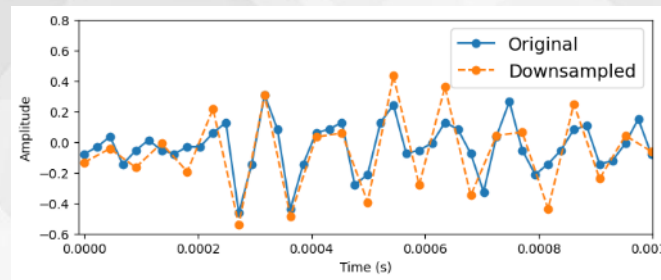
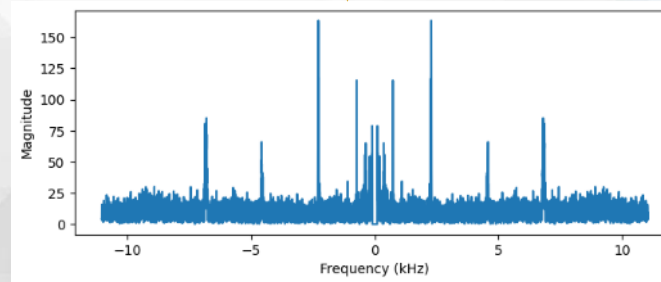
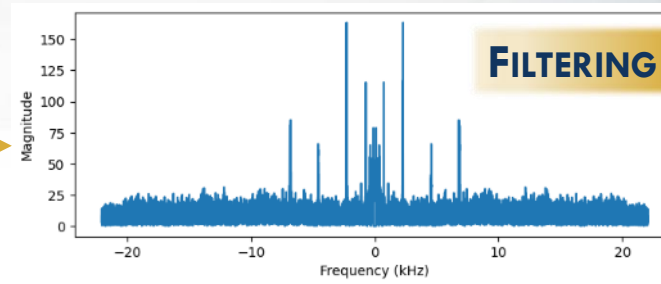
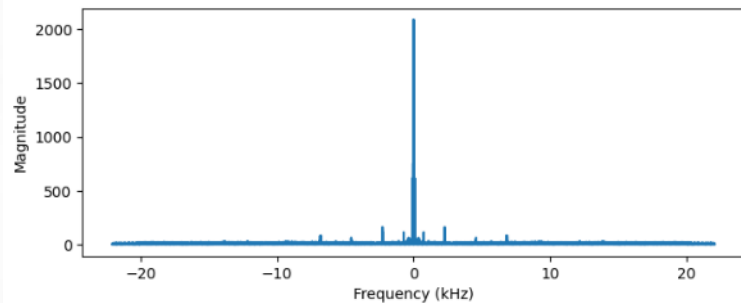
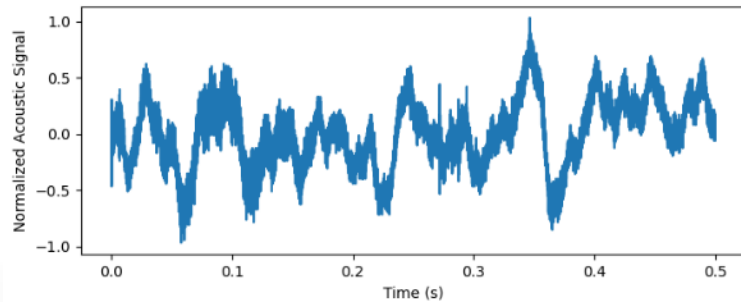
Discrete Fourier Transform (DFT)

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Inverse DFT

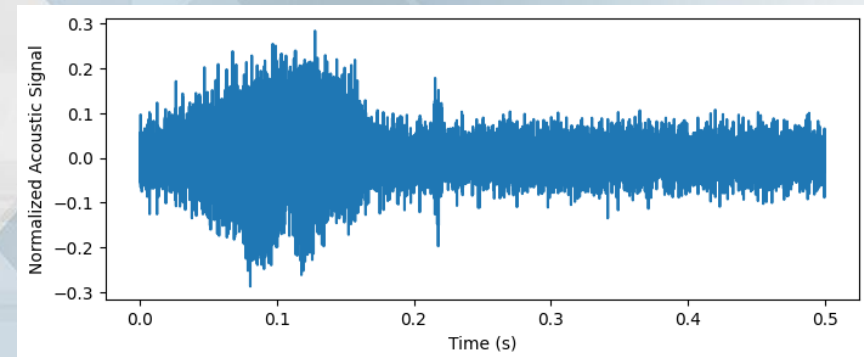
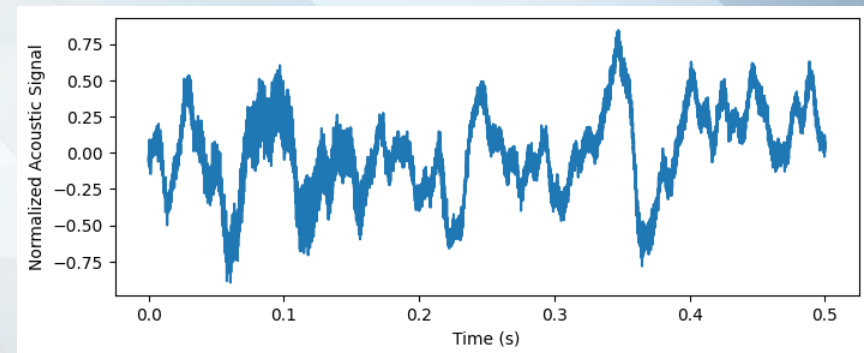
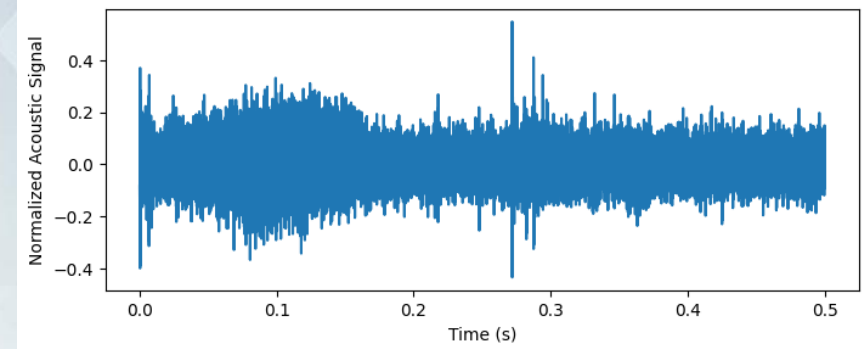
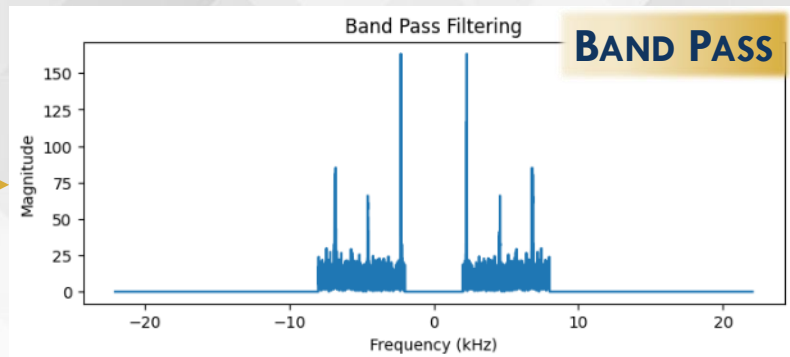
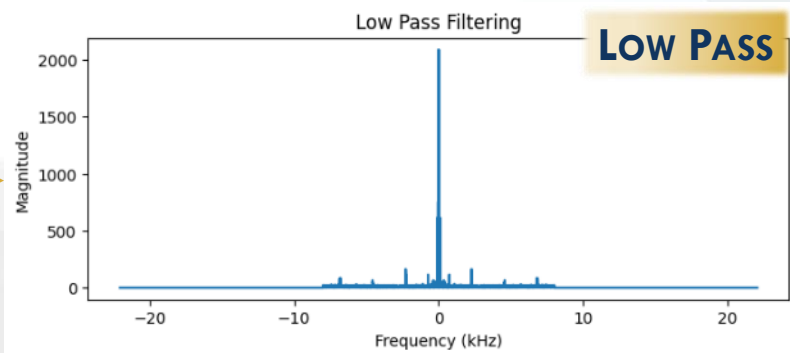
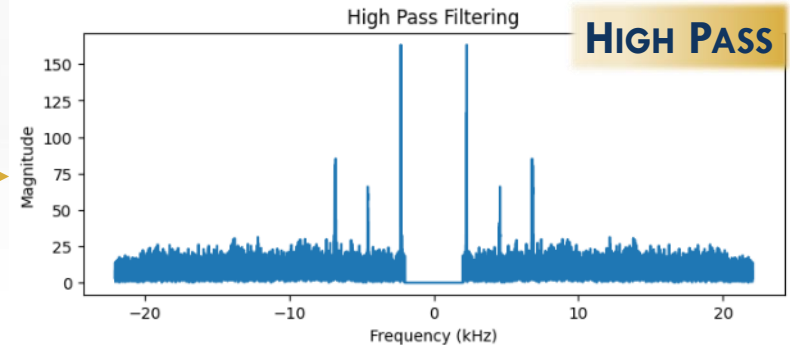
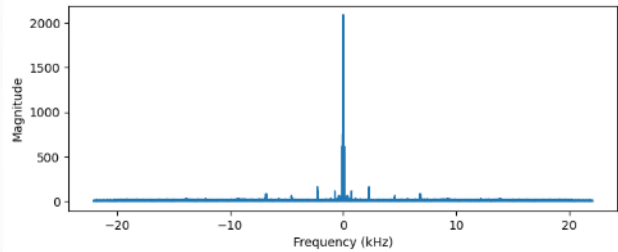
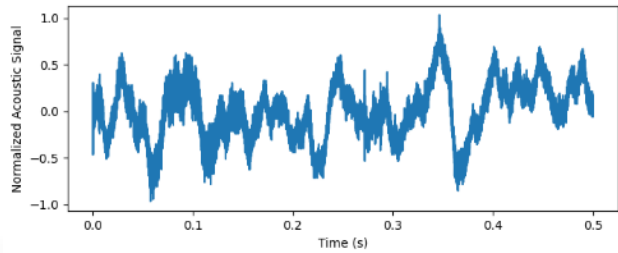
$$x[n] = \sum_{k=0}^{K-1} X[k] e^{+j2\pi \frac{nk}{K}}, n \in \{0, 1, \dots, N-1\}$$

Fourier Analysis – DSP Example

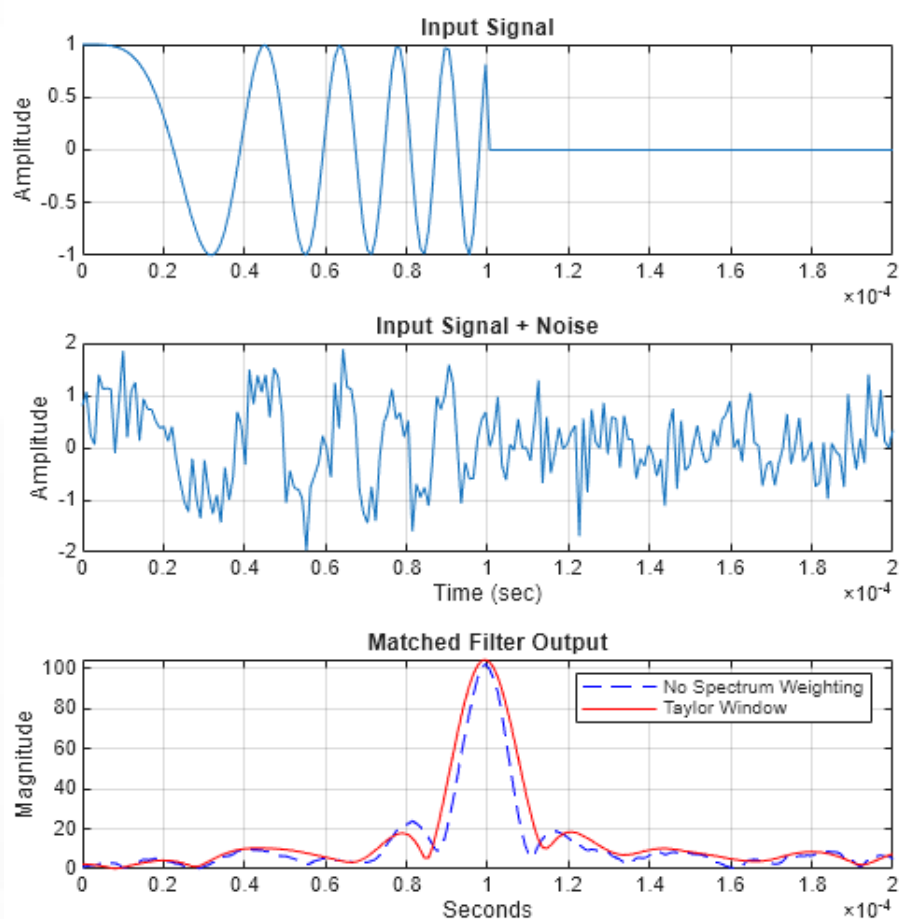


Filtering

Filtering – Basics



Filtering – Matched Filtering



- Signal-to-Noise Ratio (**SNR**):

$$\frac{|y(t_M)|^2}{n_p} = \chi = \frac{\left| \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) H(\Omega) e^{j\Omega t_M} d\Omega \right|^2}{\frac{\sigma_w^2}{2\pi} \int_{-\infty}^{\infty} |H(\Omega)|^2 d\Omega}$$

- What is the optimal choice for $H(\Omega)$?

$$\operatorname{argmax}_{H(\Omega)} \chi = \operatorname{argmax}_{H(\Omega)} \frac{\left| \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) H(\Omega) e^{j\Omega t_M} d\Omega \right|^2}{\frac{\sigma_w^2}{2\pi} \int_{-\infty}^{\infty} |H(\Omega)|^2 d\Omega}$$

<https://www.mathworks.com/help/phased/ug/matched-filtering.html>

Matched Filtering – Derivation

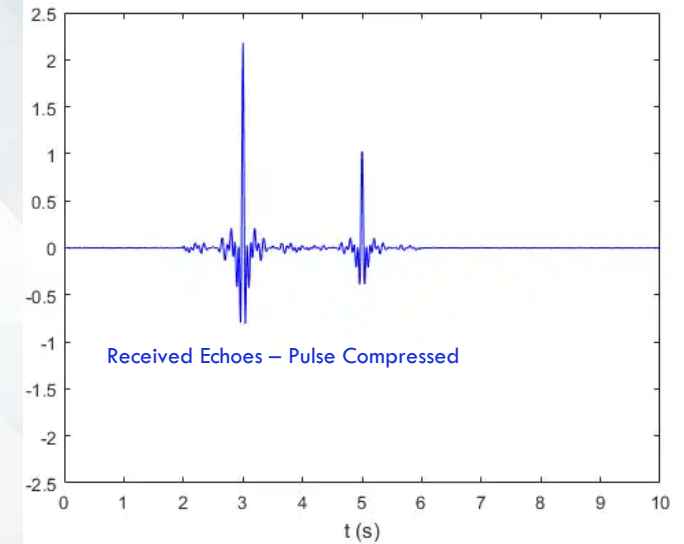
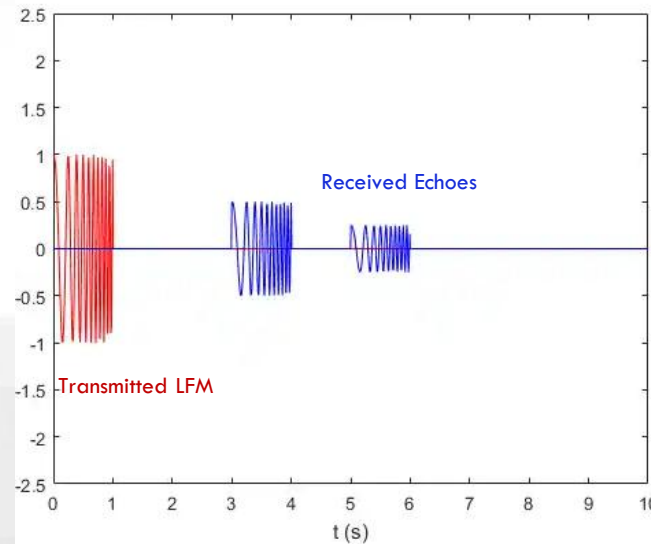
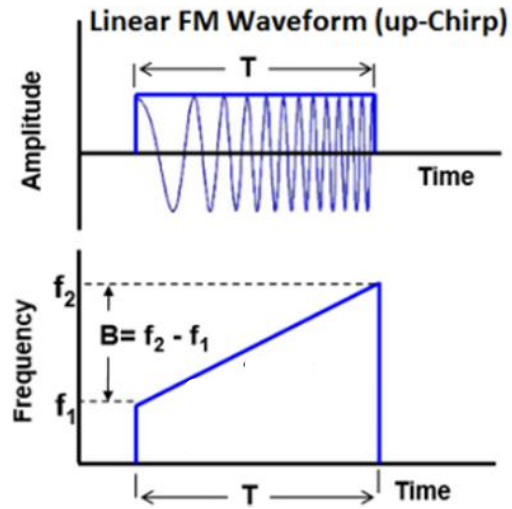
$$\chi = \frac{\left| \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) H(\Omega) e^{j\Omega t_M} d\Omega \right|^2}{\frac{\sigma_w^2}{2\pi} \int_{-\infty}^{\infty} |H(\Omega)|^2 d\Omega} \leq \frac{\frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} |X(\Omega) e^{j\Omega t_M}|^2 d\Omega \int_{-\infty}^{\infty} |H(\Omega)|^2 d\Omega}{\frac{\sigma_w^2}{2\pi} \int_{-\infty}^{\infty} |H(\Omega)|^2 d\Omega}$$

$$\rightarrow \operatorname{argmax}_{H(\Omega)} \frac{\left| \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) H(\Omega) e^{j\Omega t_M} d\Omega \right|^2}{\frac{\sigma_w^2}{2\pi} \int_{-\infty}^{\infty} |H(\Omega)|^2 d\Omega} \rightarrow H_M(\Omega) = \alpha X^*(\Omega) e^{-j\Omega t_M}$$

$$\max_{H(\Omega)} \chi = \frac{\left| \frac{\alpha}{2\pi} \int_{-\infty}^{\infty} |X(\Omega)|^2 d\Omega \right|^2}{\frac{|\alpha|^2 \sigma_w^2}{2\pi} \int_{-\infty}^{\infty} |X(\Omega)|^2 d\Omega} = \frac{1}{2\pi \sigma_w^2} \int_{-\infty}^{\infty} |X(\Omega)|^2 d\Omega = \frac{\text{Energy}(x(t))}{2\pi \sigma_w^2}$$

Matched Filtering – Pulse Compression

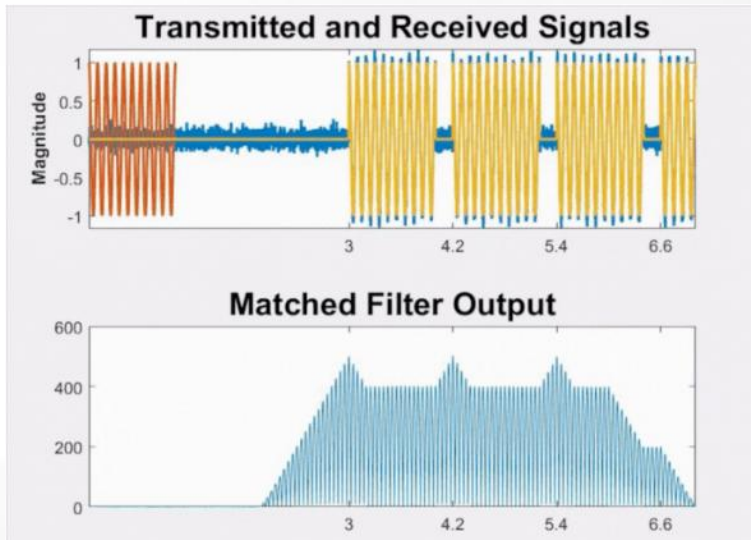
Chirp Sonar Pulses + Matched Filtering → Pulse Compression



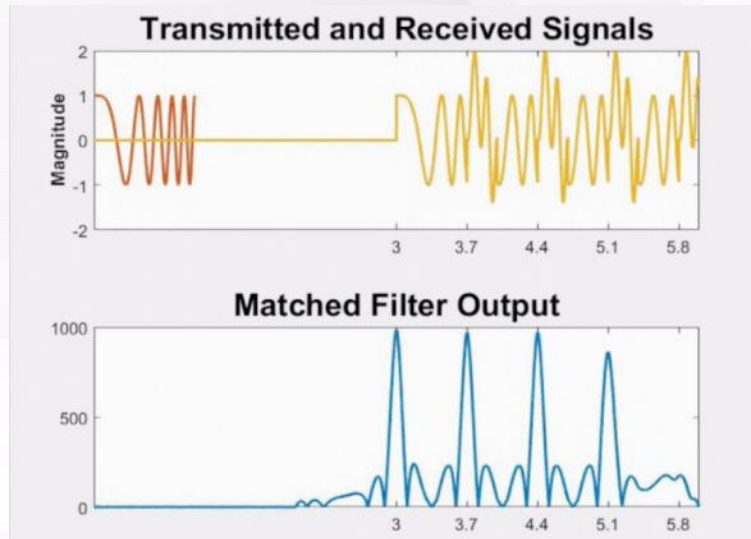
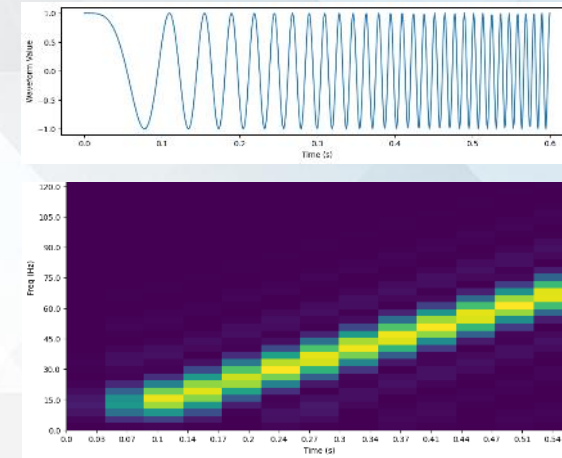
https://en.wikipedia.org/wiki/Pulse_compression



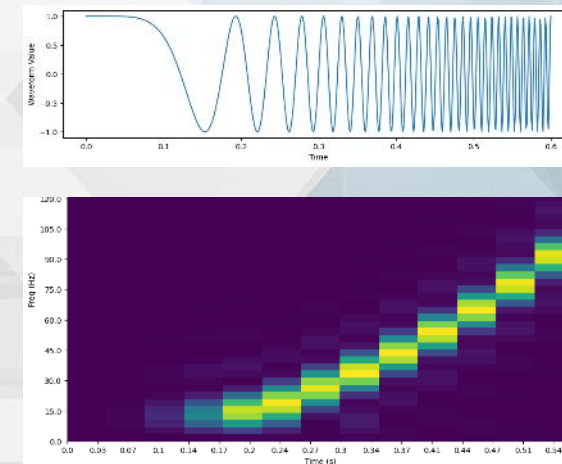
Matched Filtering – Projected Pulse Types



Linear Frequency Modulation (Up)

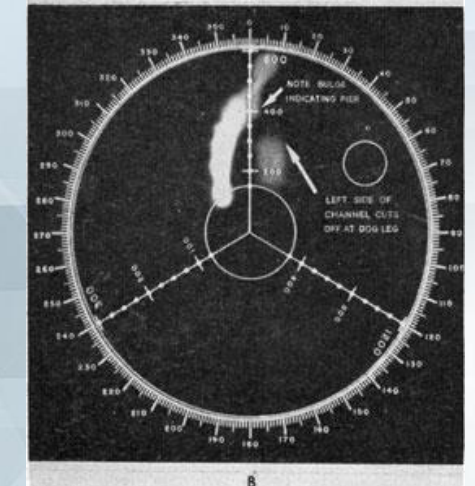
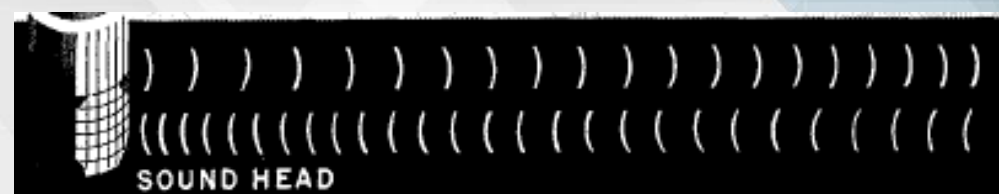


Quadratic Frequency Modulation (Up)



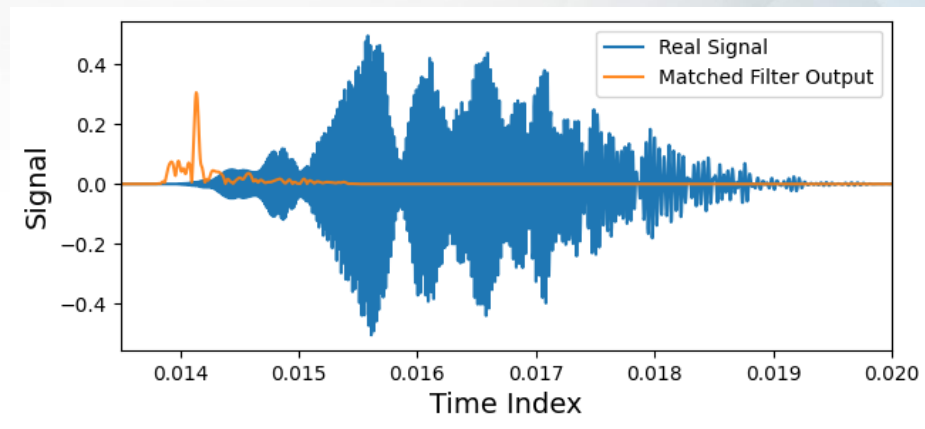
D. N. Saikia, 2024.

NSWC PANAMA CITY DIVISION



20

Filtering – Notebook Example

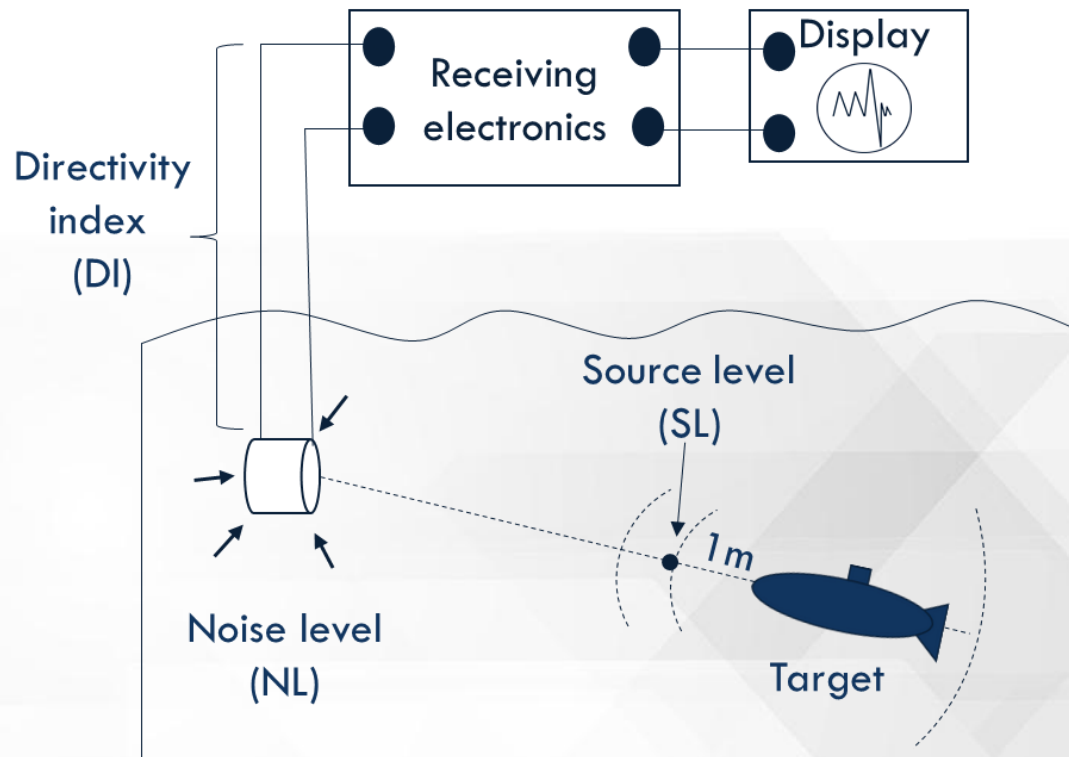


COFFEE BREAK

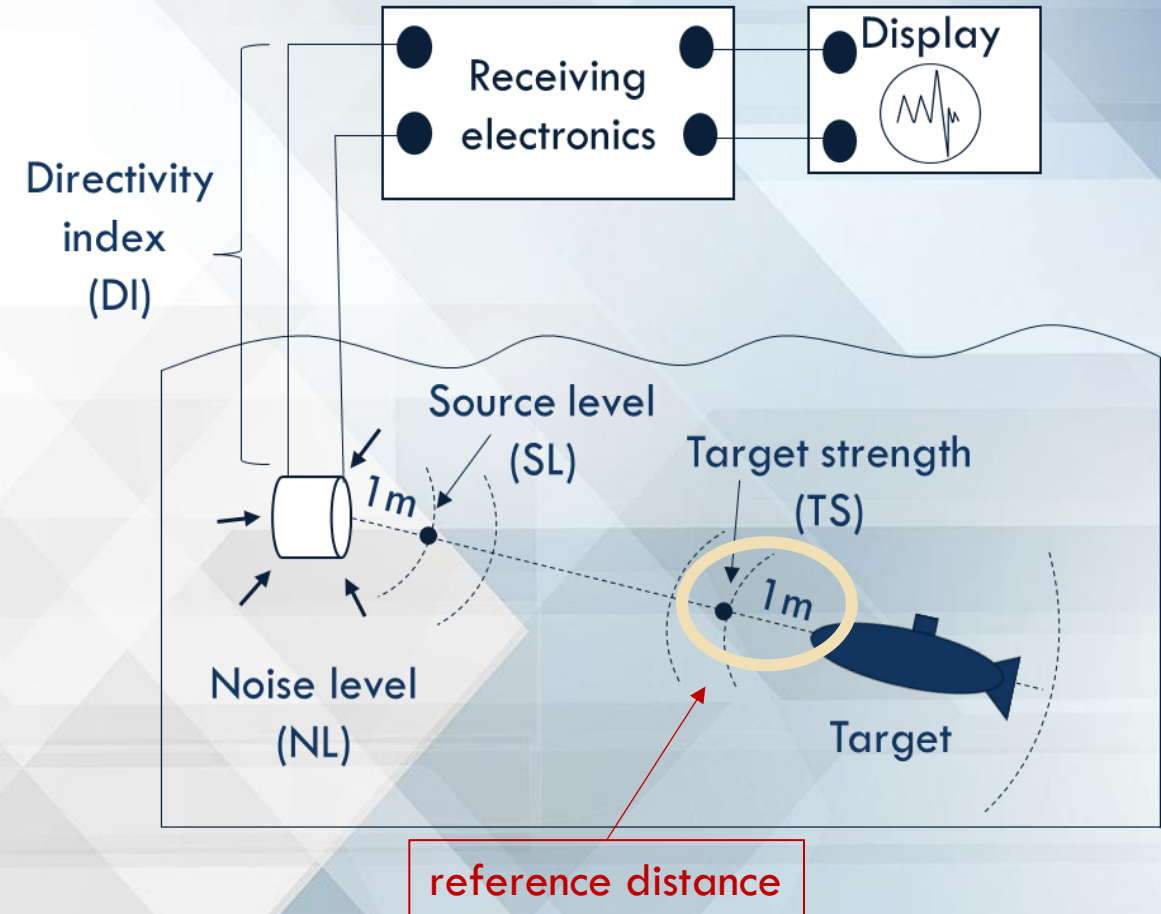


The Sonar Equation – Single-element, Multipath, and Noise

Passive Sonar



Active Sonar



The Sonar Equation

Sonar equations are based on the equality between the desired(i.e., the signal) and the undesired(i.e., the background) portions of the received signal

$$\text{signal} = \text{background}$$

Passive Sonar (only listen)

$$SL - TL = NL - DI$$

$$SNR_{\text{pass}} = SL - TL - (NL - DI)$$

Active Sonar (project and listen)

$$SL - 2TL + TS = RL \oplus (NL - DI)$$

$$SNR_{\text{act}} = SL - 2TL + TS - RL \oplus (NL - DI)$$

SL – Source Level

TL – Transmission Loss

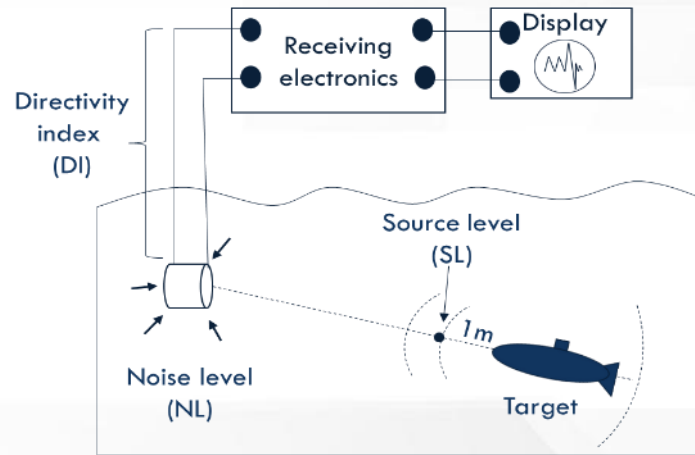
NL – Noise Level (ambient)

DI – Directivity Index

TS – Target Strength

RL – Reverberation Level

The Sonar Equation



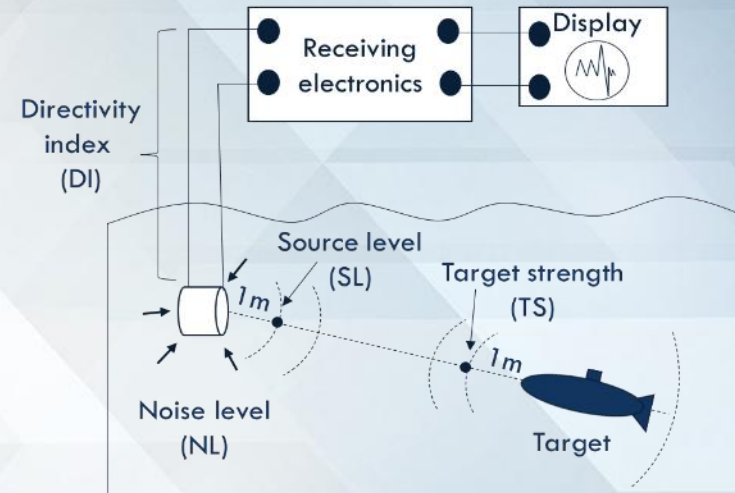
radiated noise of the target at 1 m

assuming that the background is isotropic noise; different for reverberation

$$SL - TL - (NL - DI)$$

signal reaching hydrophone is reduced by transmission loss

background is reduced by directivity index



source level at 1m along hydrophone's axis

signal attenuated on the way back to source

assuming that the background is isotropic noise; different for reverberation

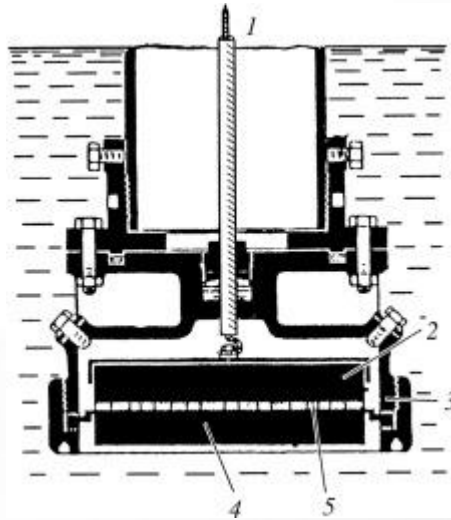
$$SL - TL + TS - TL - (NL - DI)$$

signal reaching target is reduced by transmission loss

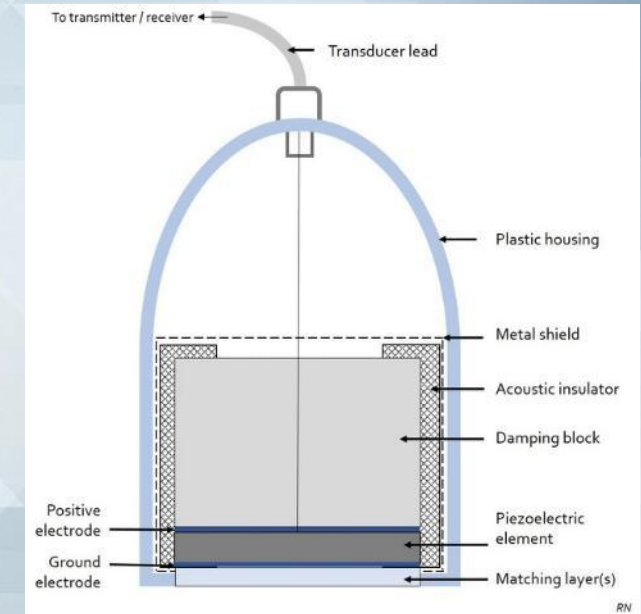
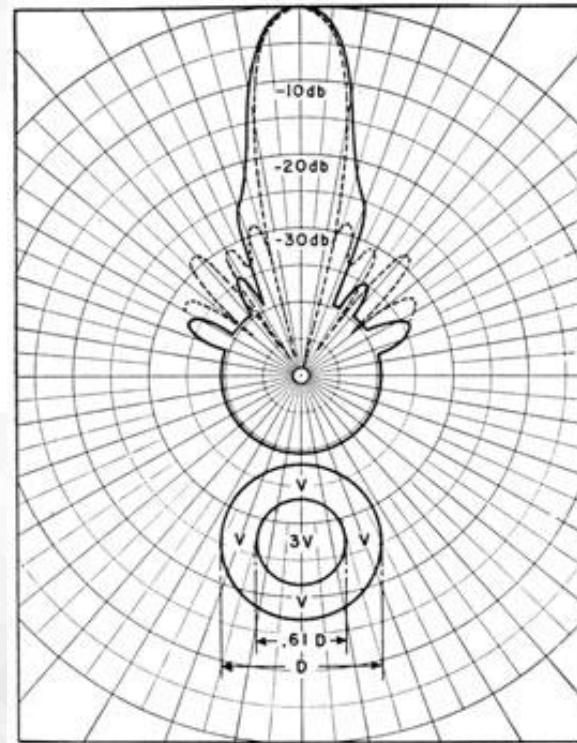
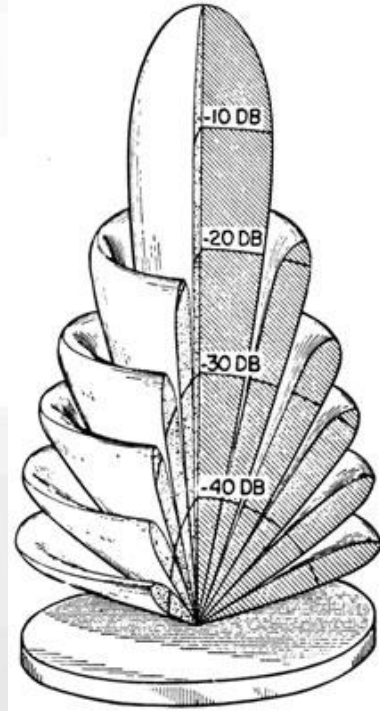
signal reflected or backscattered (by target) towards source

background is reduced by directivity index

The Sonar Equation – Transmission & Reception



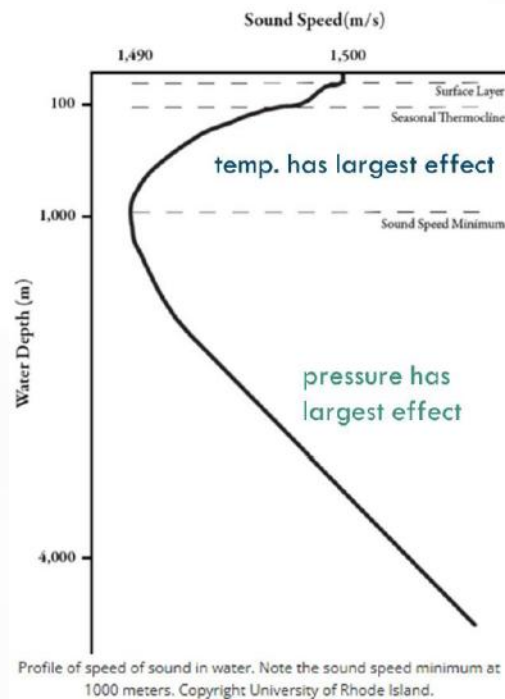
5 – Quartz Material
2 and 4 – Inner and Outer Electrodes, resp.



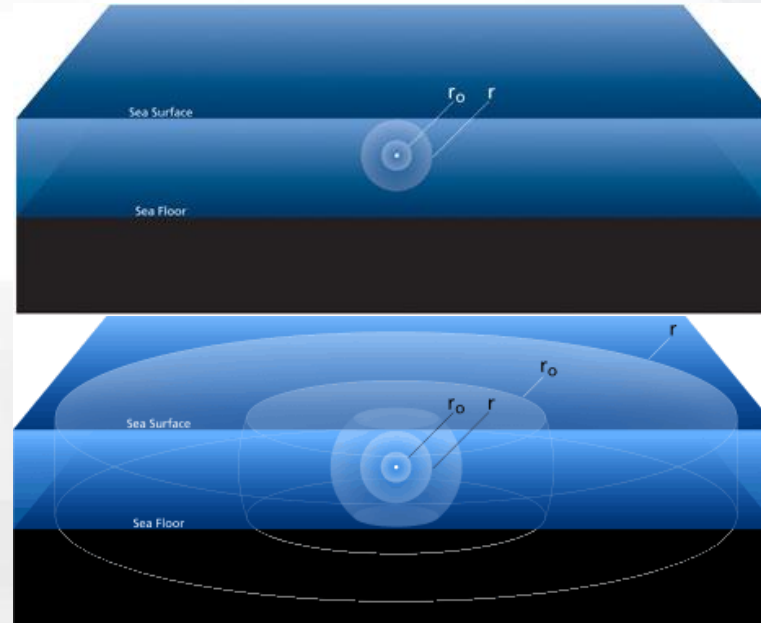
1. B. Bensaude-Vincent and F. Duck, Paul Langevin: Physicist and Social Activist 2025.
2. Naval Sonar, 1953, Available: <https://maritime.org/doc/sonar/>.
3. R. Nightingale et al., "Ultrasound transducer," Radiopaedia.org, 2026, doi: 10.53347/rID-14565.

The Sonar Equation – Signal Propagation

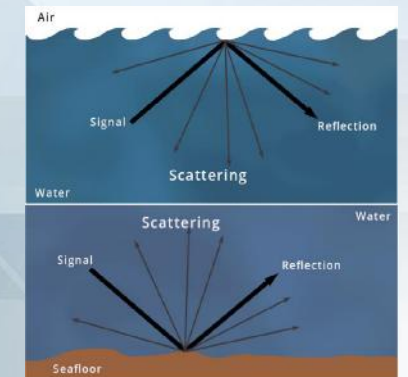
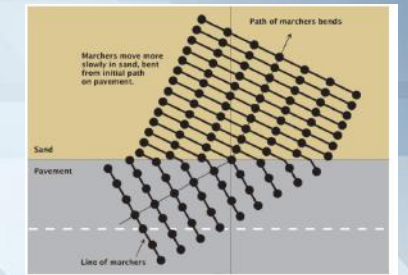
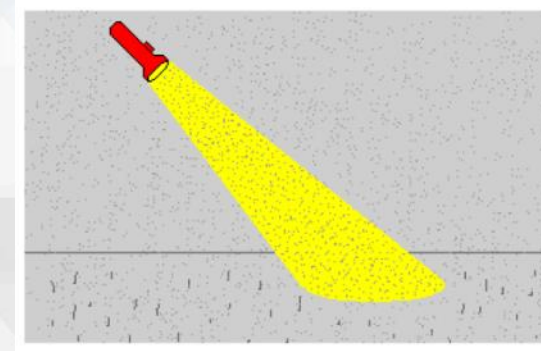
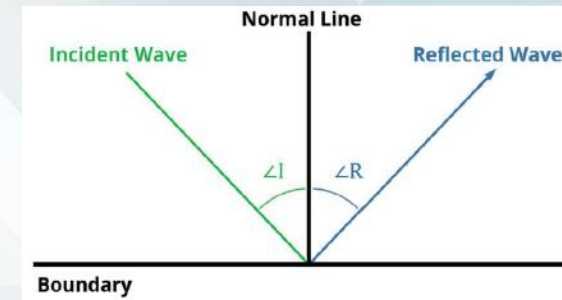
Sound Speed Profile



Spreading Losses

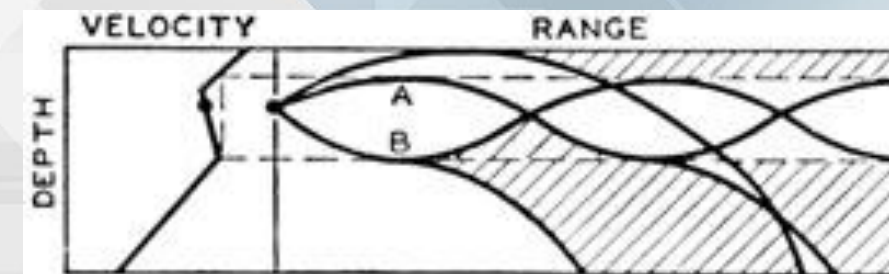
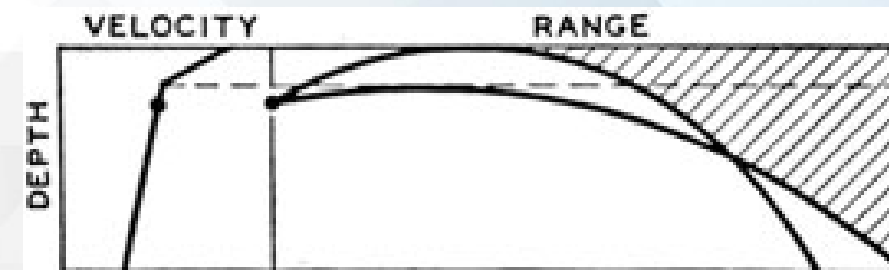
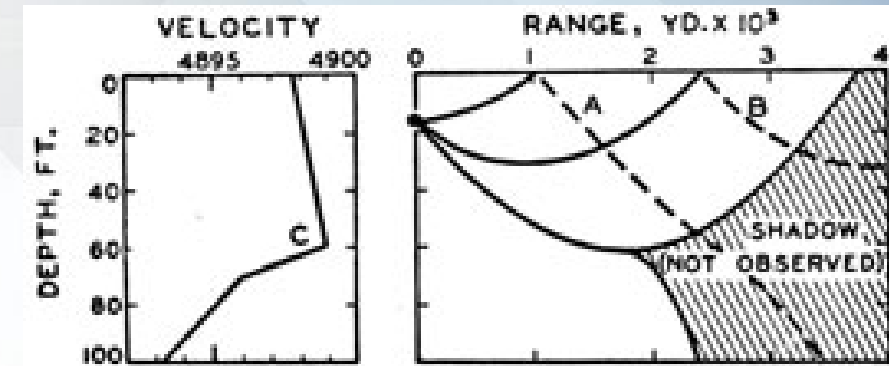
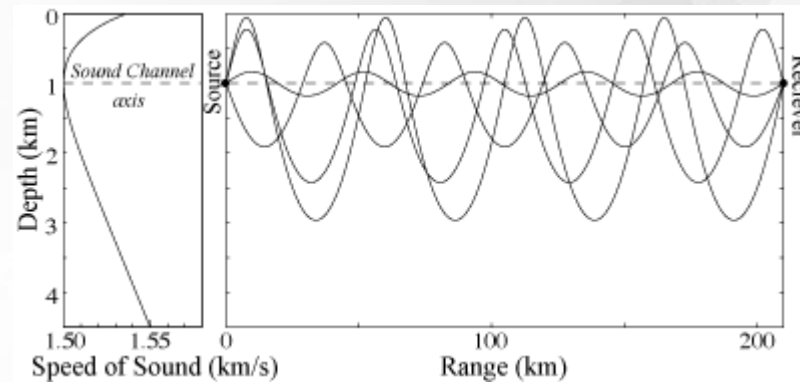
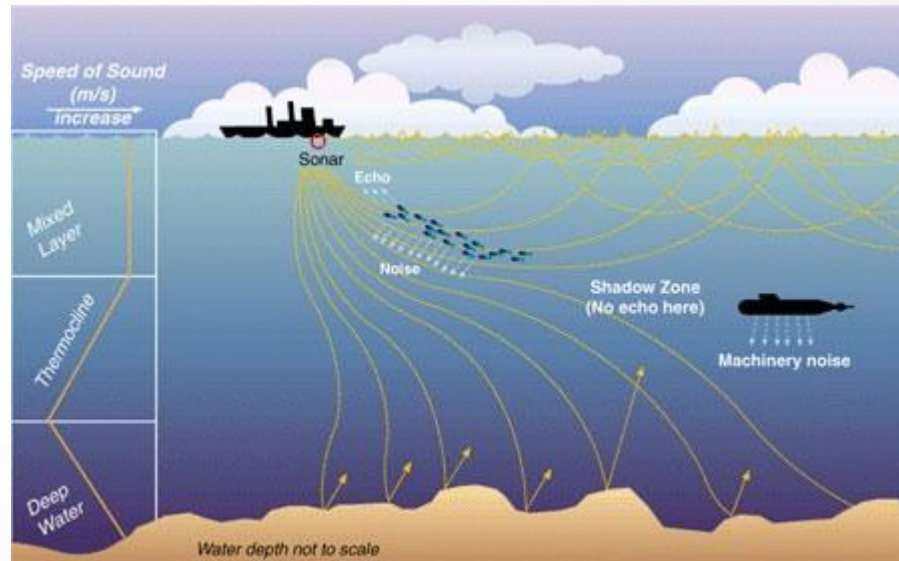


Reflection, Refraction, Scattering, Reverberation



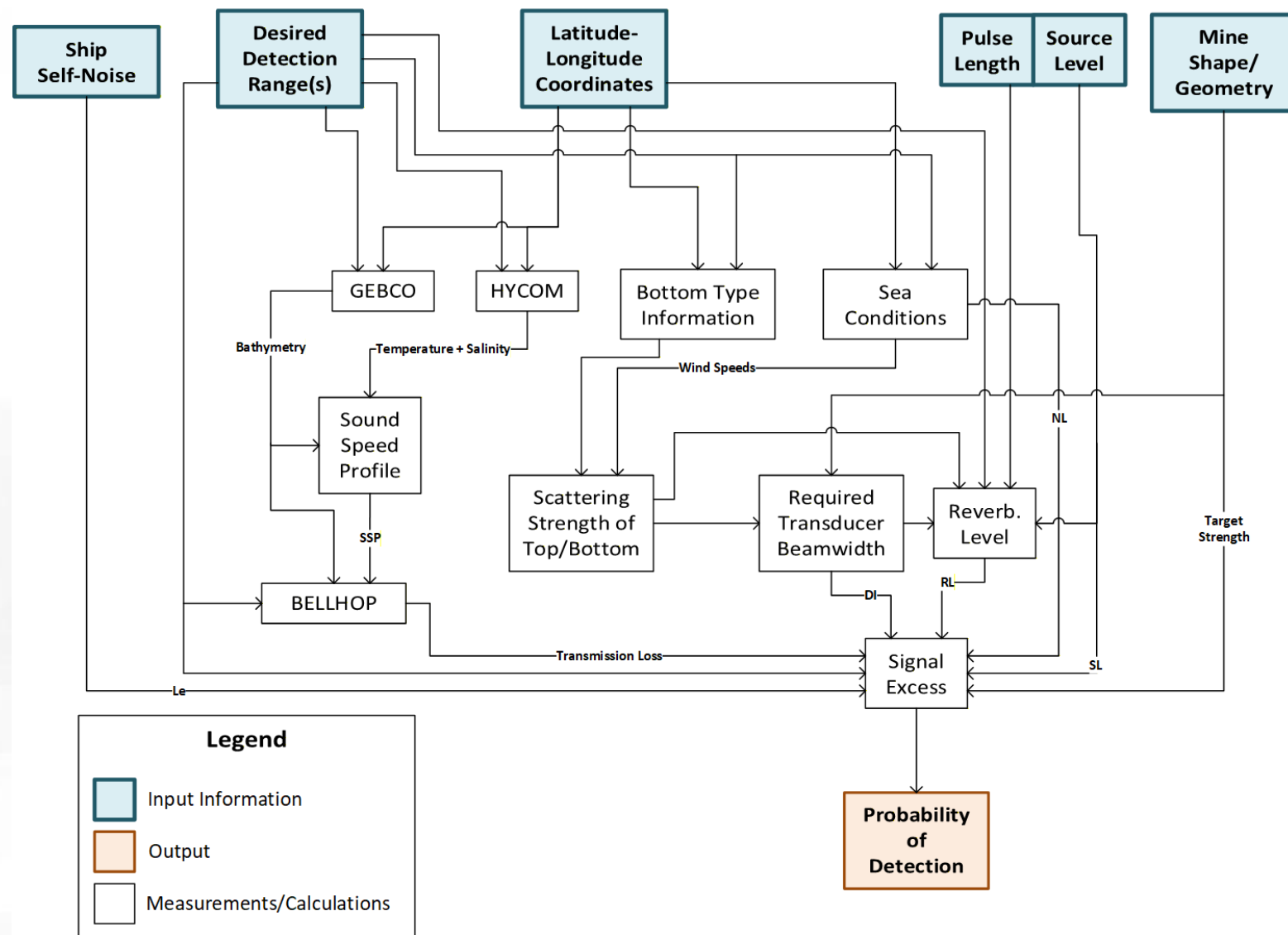
Images from Discovery of Sound in the Sea at <https://dosits.org/science>.

The Sonar Equation – Signal Propagation

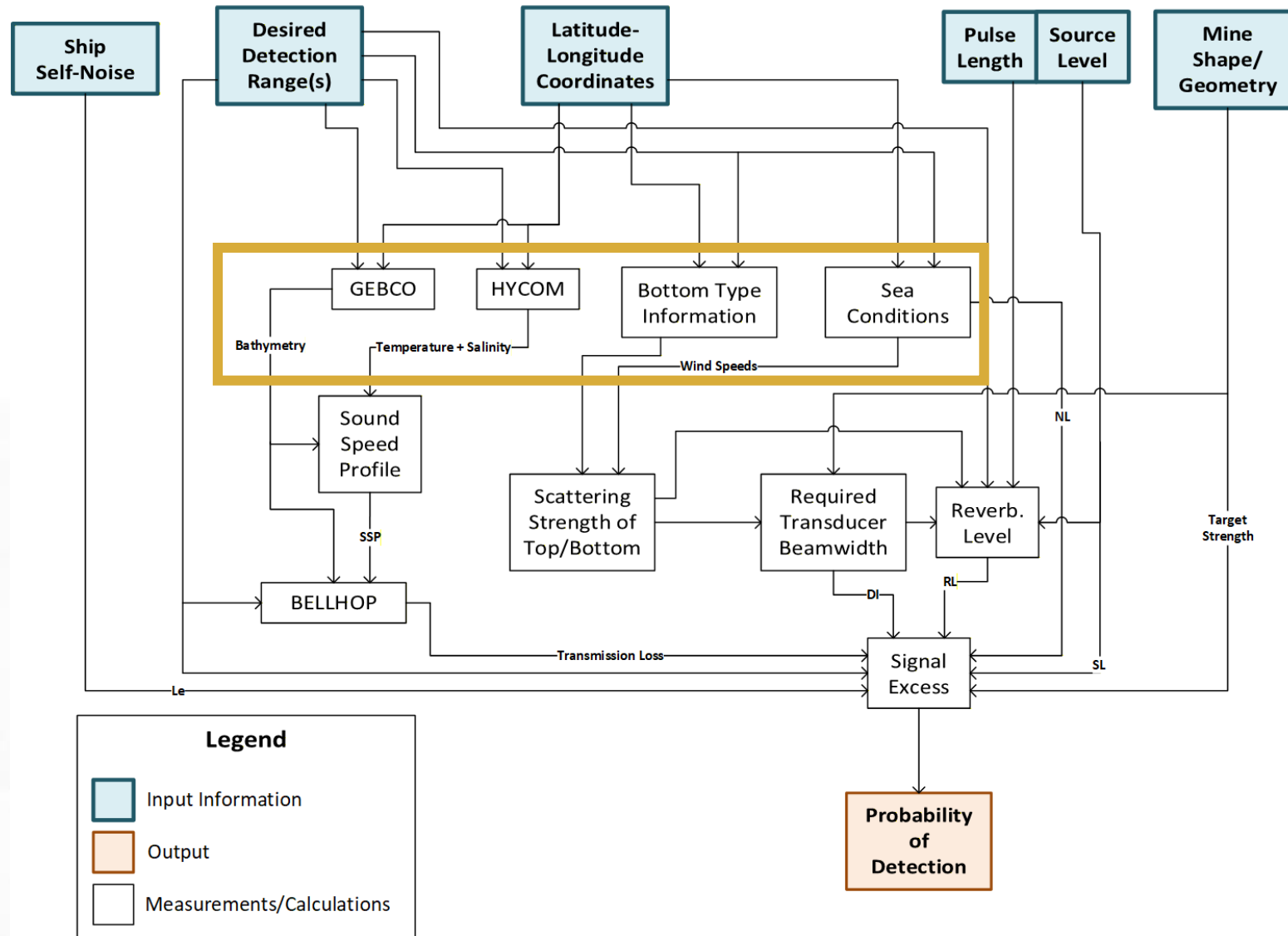


Images from Discovery of Sound in the Sea at <https://dosits.org/science> and "Naval Sonar", Ch. 1.

Flowchart for Sonar Equation Calculations

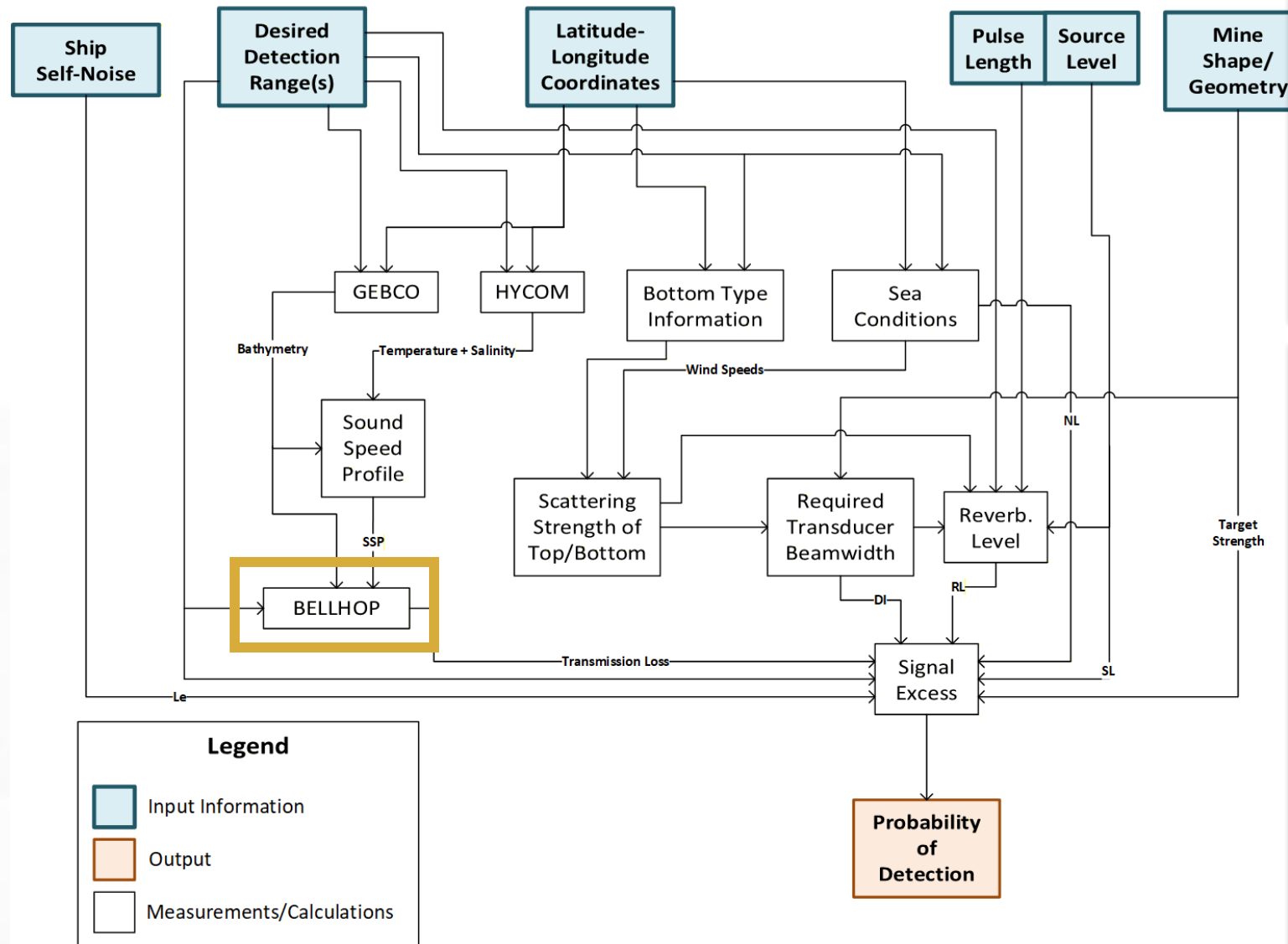


Flowchart for Sonar Equation Calculations

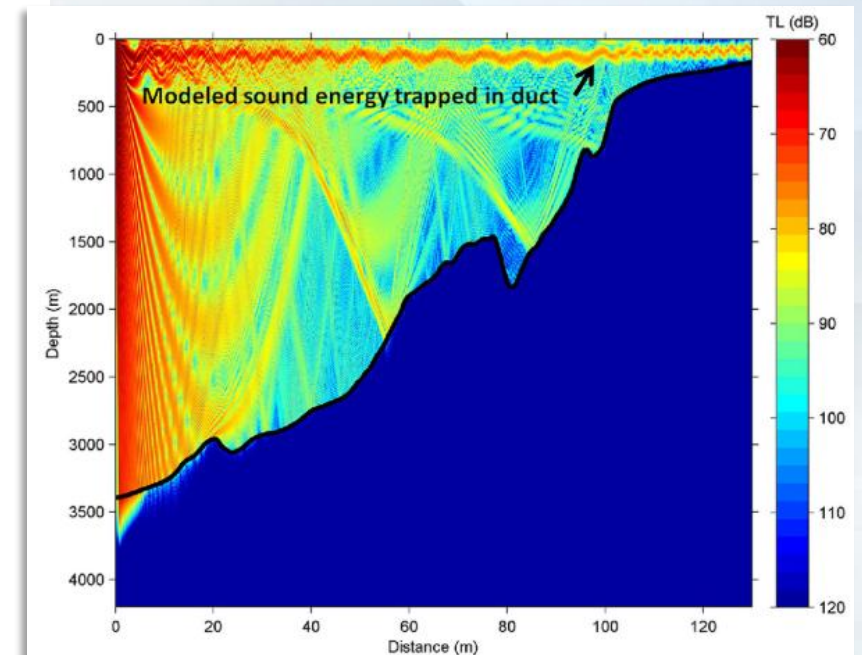
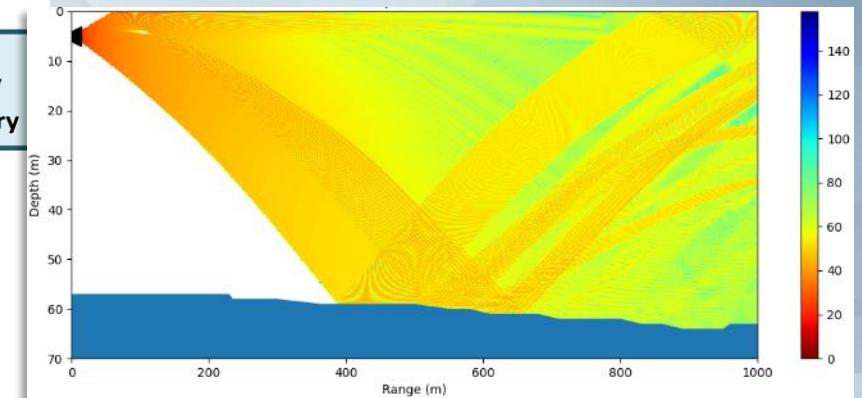


HYBRID COORDINATE
HYCOM
 OCEAN MODEL

Flowchart for Sonar Equation Calculations



BELLHOP Simulation



Visualization of parabolic equation modeling. The color bar shows transmission loss (TL) in terms of decibels (dB). Figure from Lynch, J.F., G.G. Gawarkiewicz, Y.-T. Lin, T.F. Duda, and A.E. Newhall. 2018. Impacts of ocean warming on acoustic propagation over continental shelf and slope regions. *Oceanography* 31(2):174-181, <https://doi.org/10.5670/oceanog.2018.219>.

Flowchart for Sonar Equation Calculations

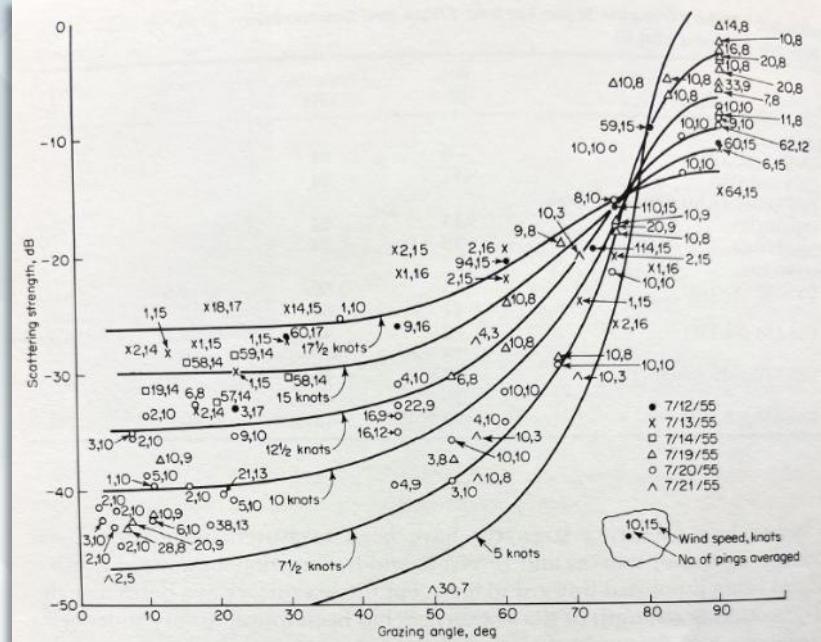
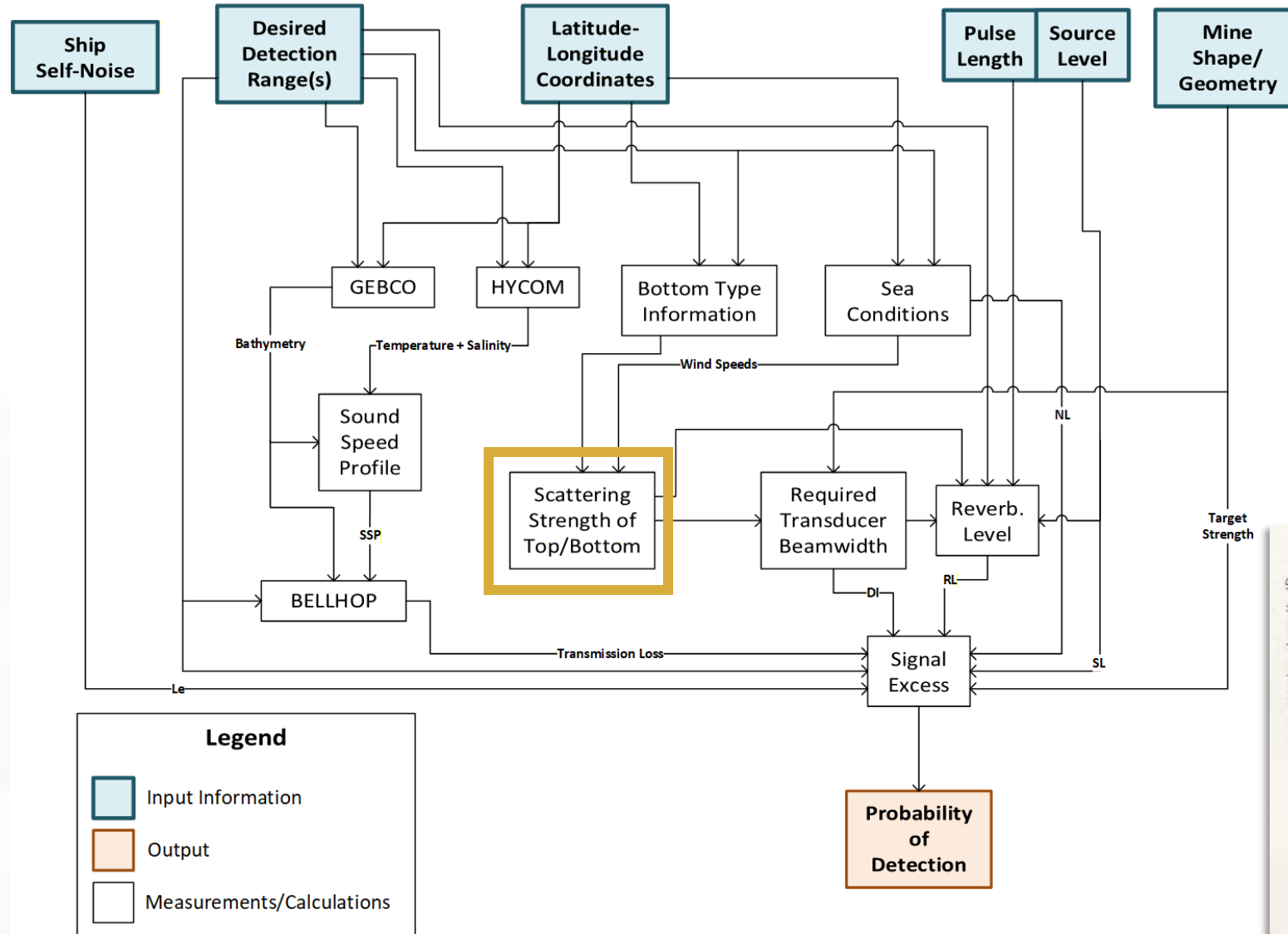


fig. 8.18 Variation of sea-surface scattering strength at 60 kHz with angle at different wind speeds off Key West, Florida. Plotted points show the number of pulses averaged and the wind speed for individual determinations. (Ref. 5.)

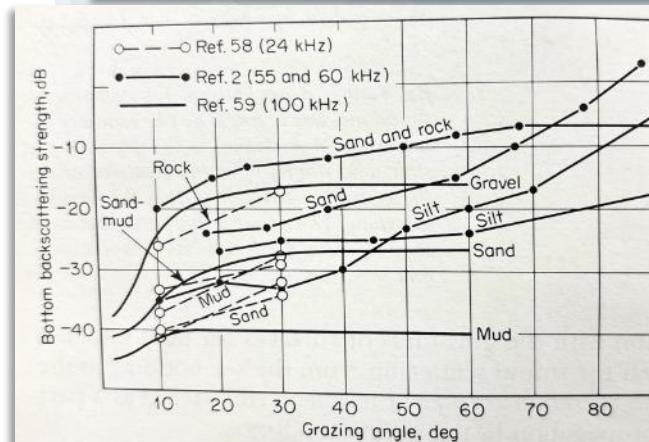


fig. 8.27 Measured backscattering strength of the seabed at various coastal locations.

Urick, 1983.

Flowchart for Sonar Equation Calculations

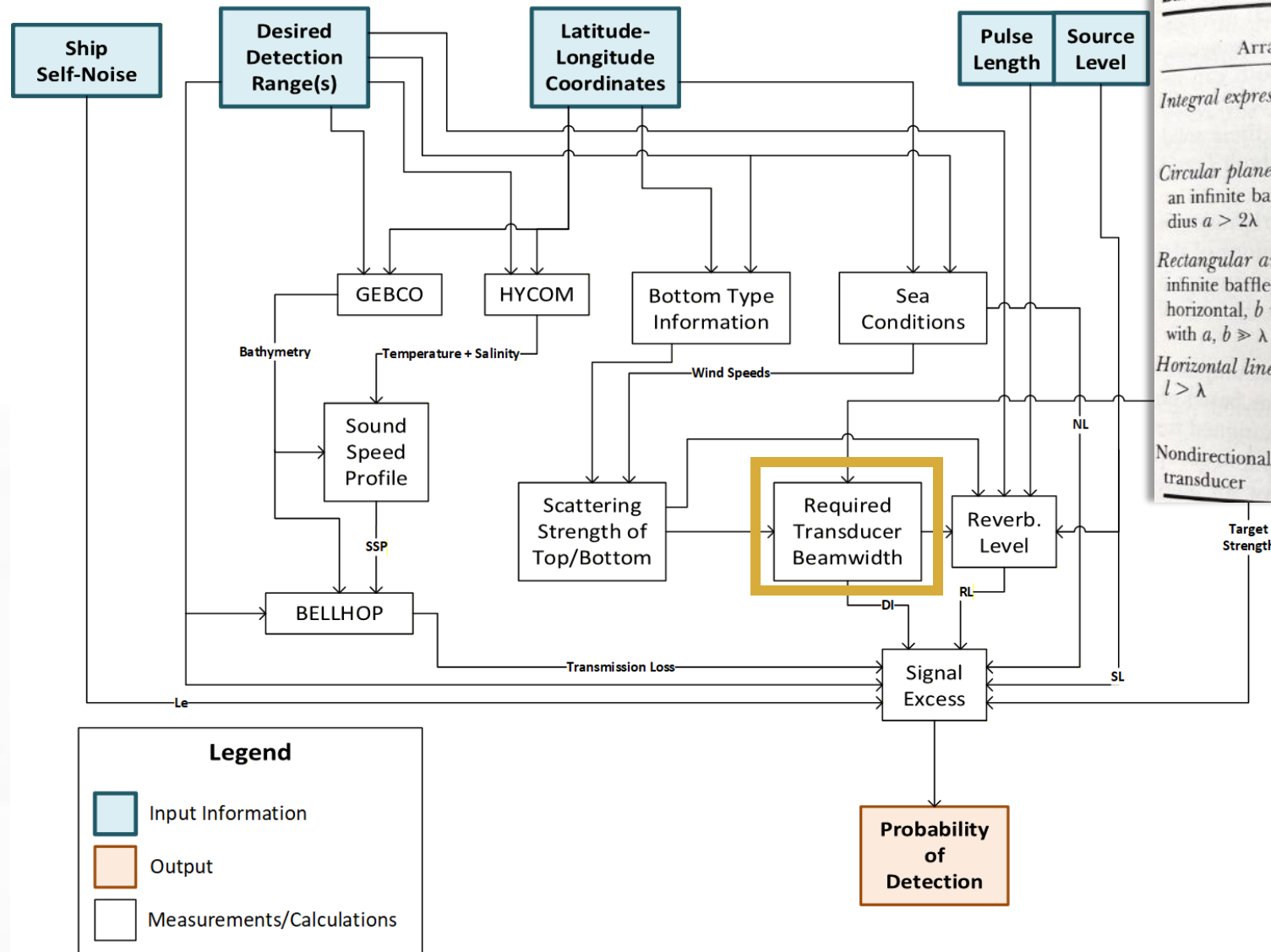
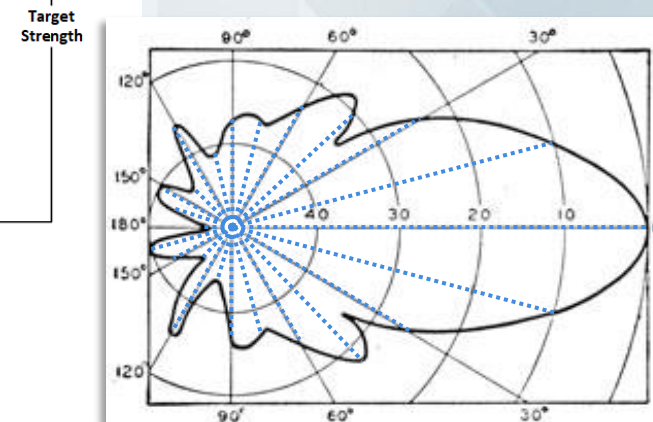


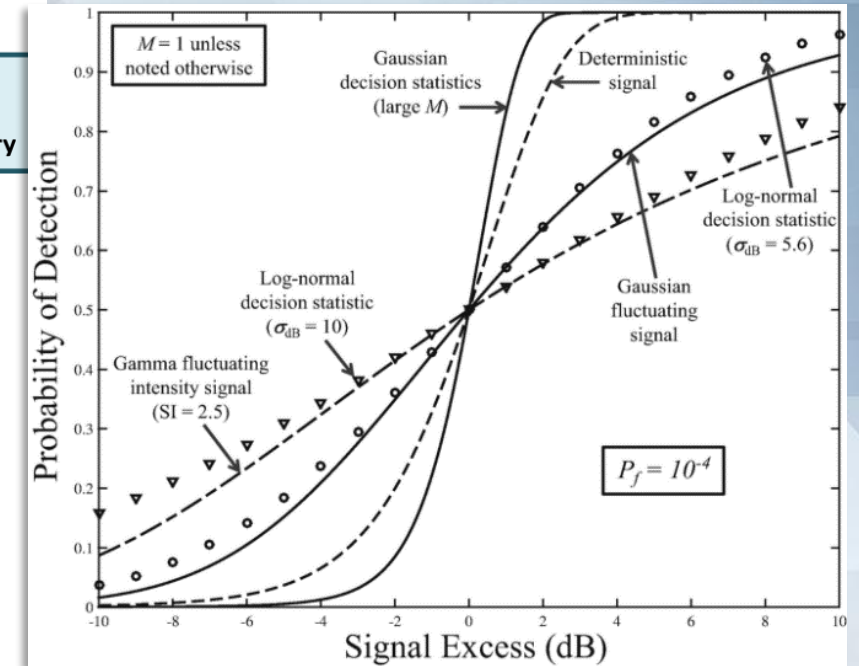
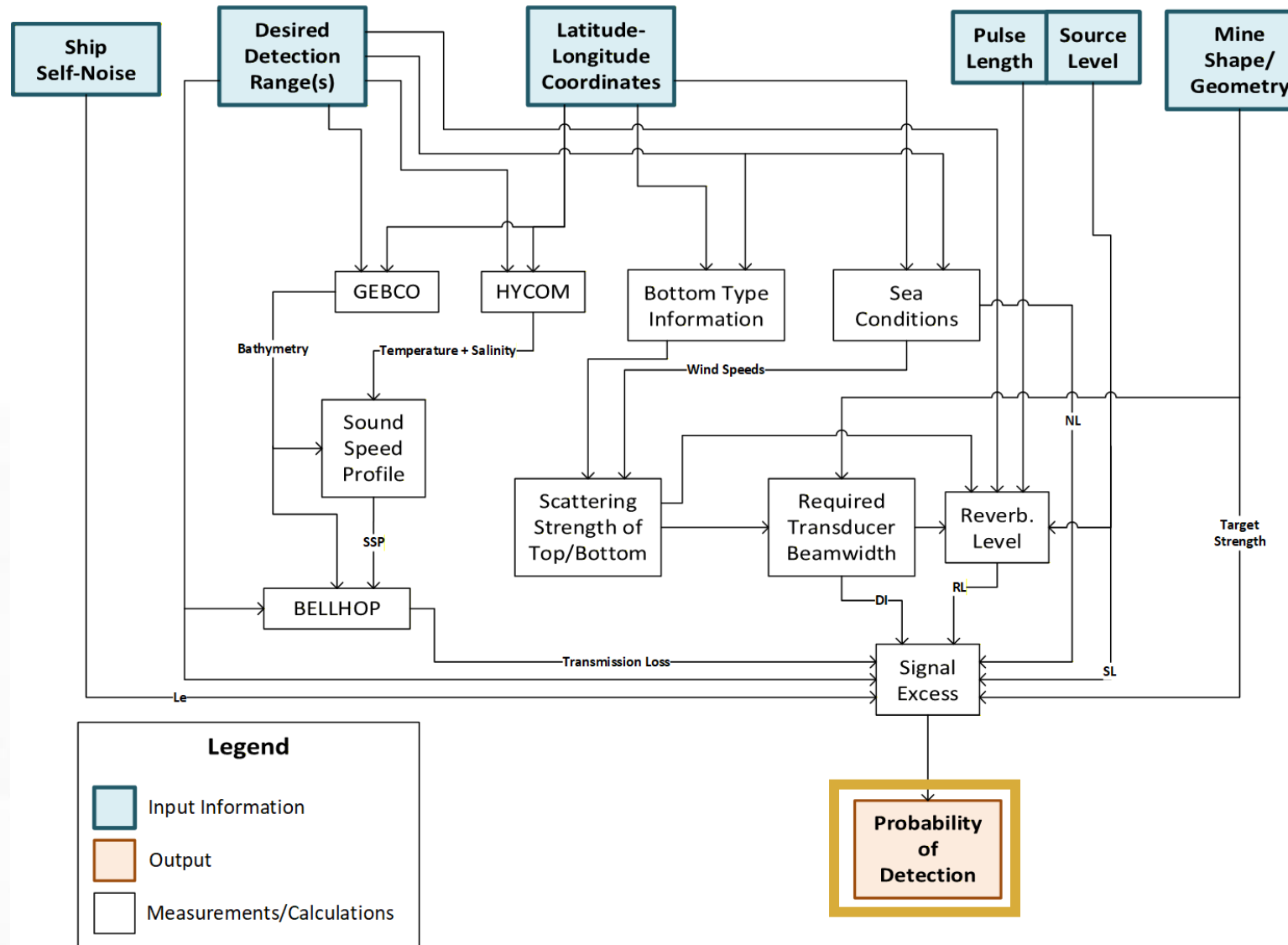
table 8.1 Equivalent Two-Way Beam Widths in Logarithmic Units, Based on Ref. 92

Array	$10 \log \Psi$ dB re 1 steradian	$10 \log \Phi$ dB re 1 radian
<i>Integral expression</i>	$10 \log \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} b(\theta, \varphi) b'(\theta, \varphi) \times \cos \theta d\theta d\varphi$	$10 \log \int_0^{2\pi} b(\theta, \varphi) b'(\theta, \varphi) \cdot d\varphi$
<i>Circular plane array, in an infinite baffle of radius $a > 2\lambda$</i>	$20 \log \left(\frac{\lambda}{2\pi a} \right) + 7.7$ or $20 \log y - 31.6$	$10 \log \frac{\lambda}{2\pi a} + 6.9$ or $10 \log y - 12.8$
<i>Rectangular array in an infinite baffle, side a horizontal, b vertical, with $a, b \gg \lambda$</i>	$10 \log \frac{\lambda^2}{4\pi ab} + 7.4$ or $10 \log y_a y_b - 31.6$	$10 \log \frac{\lambda}{2\pi a} + 9.2$ or $10 \log y_a - 12.6$
<i>Horizontal line of length $l > \lambda$</i>	$10 \log \frac{\lambda}{2\pi l} + 9.2$ or $10 \log y - 12.8$	$10 \log \frac{\lambda}{2\pi l} + 9.2$ or $10 \log y - 12.8$
<i>Nondirectional (point) transducer</i>	$10 \log 4\pi = 11.0$	$10 \log 2\pi = 8.0$

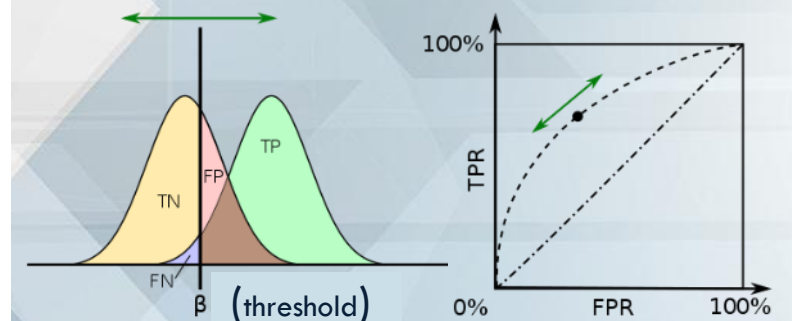


[1] Urick, 1983.
[2] Naval Sonar, 1953.

Flowchart for Sonar Equation Calculations



Abraham, D.A. (2019). Sonar Systems and the Sonar Equation. In: Underwater Acoustic Signal Processing. Modern Acoustics and Signal Processing. Springer, Cham.
https://doi.org/10.1007/978-3-319-92983-5_2

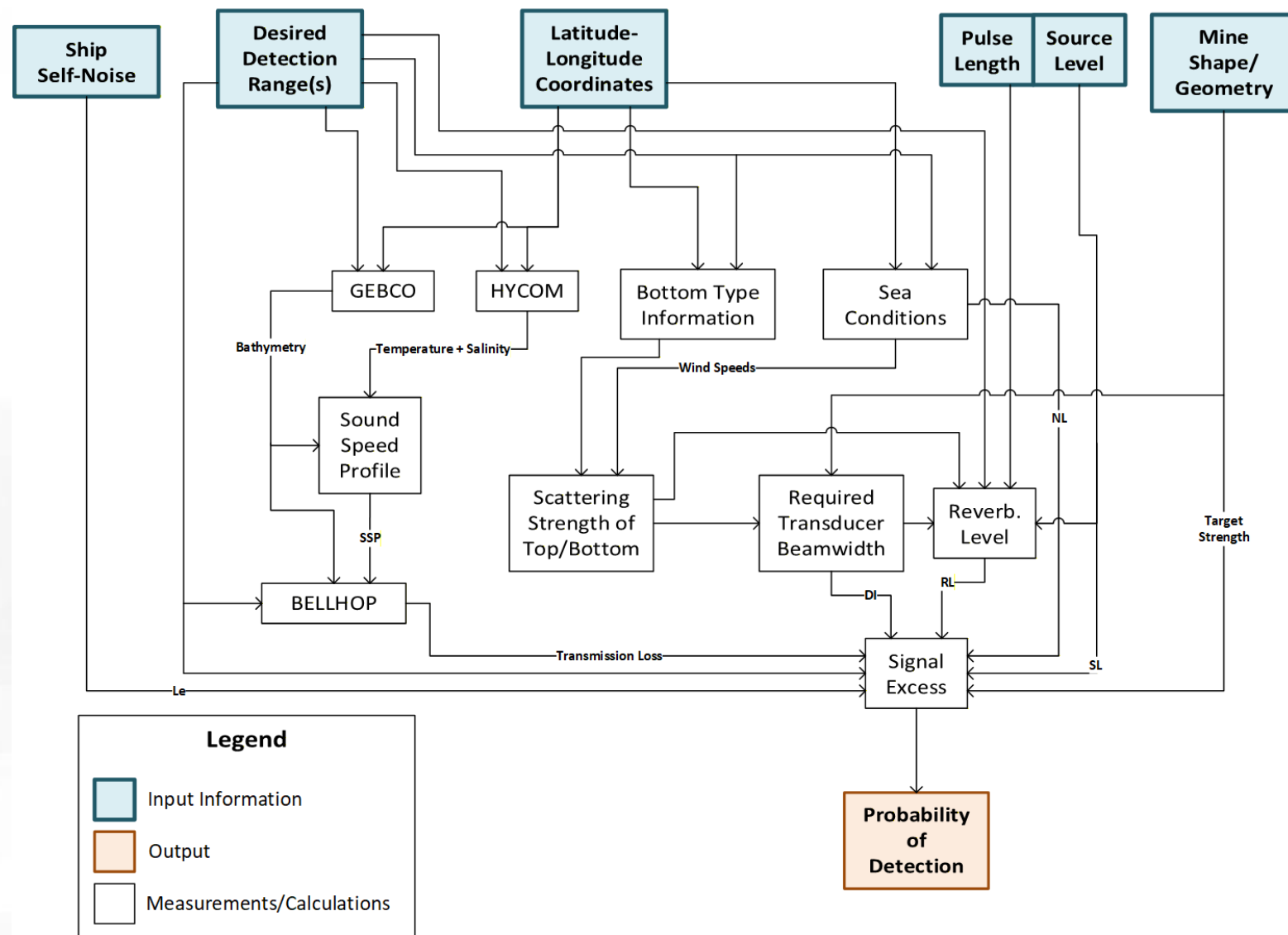


Sonar Eqn

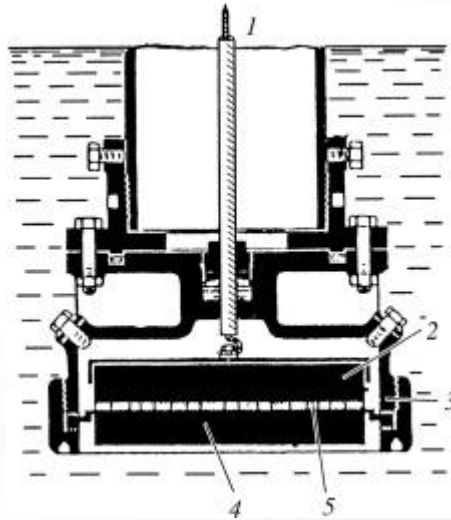
Beamforming

Sidescan/SAS

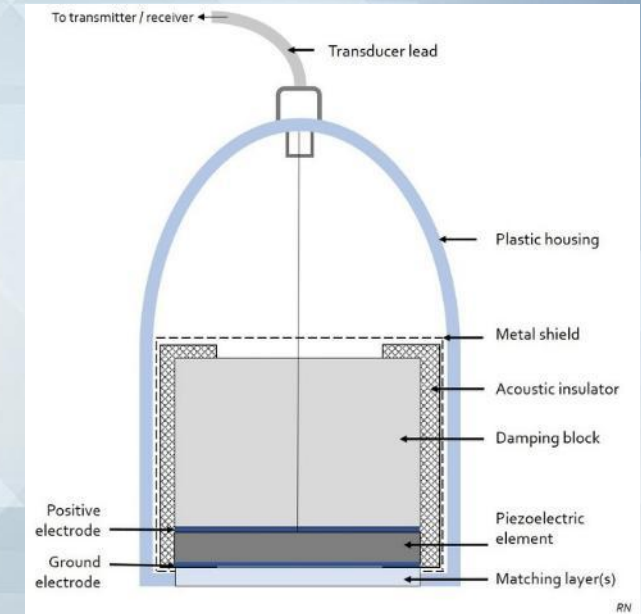
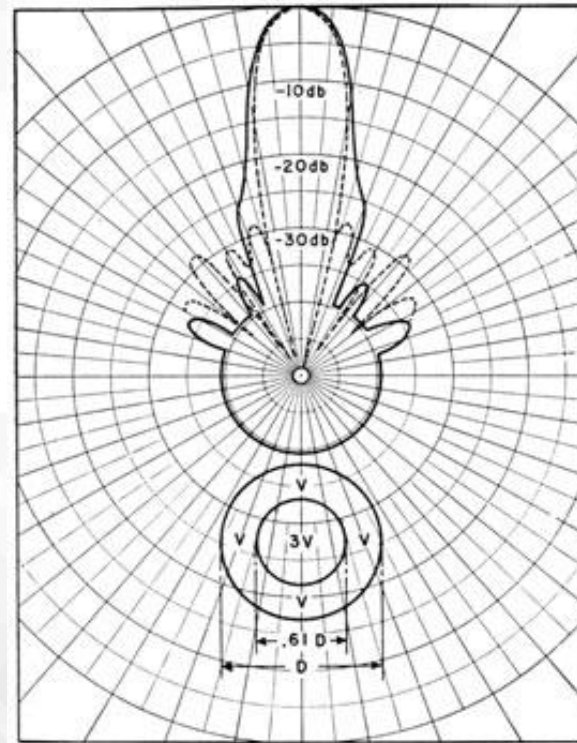
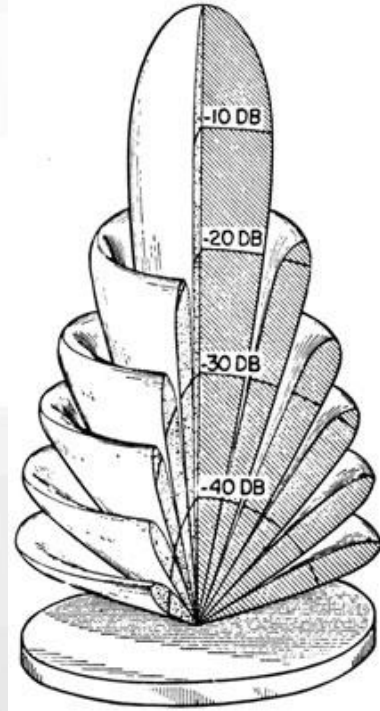
Flowchart for Sonar Equation Calculations



Beam Patterns and Beamforming

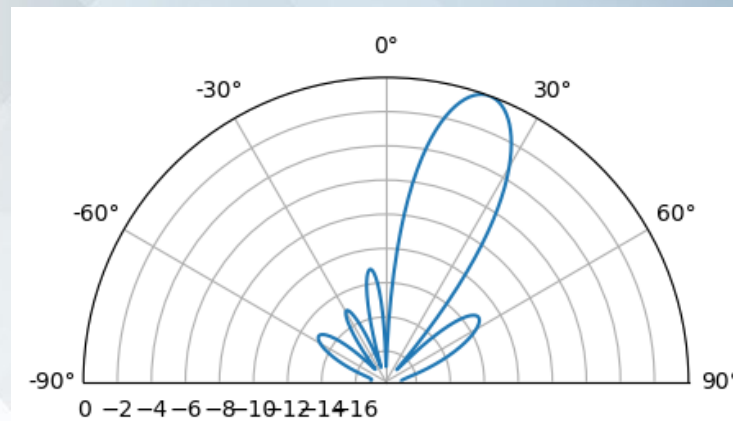
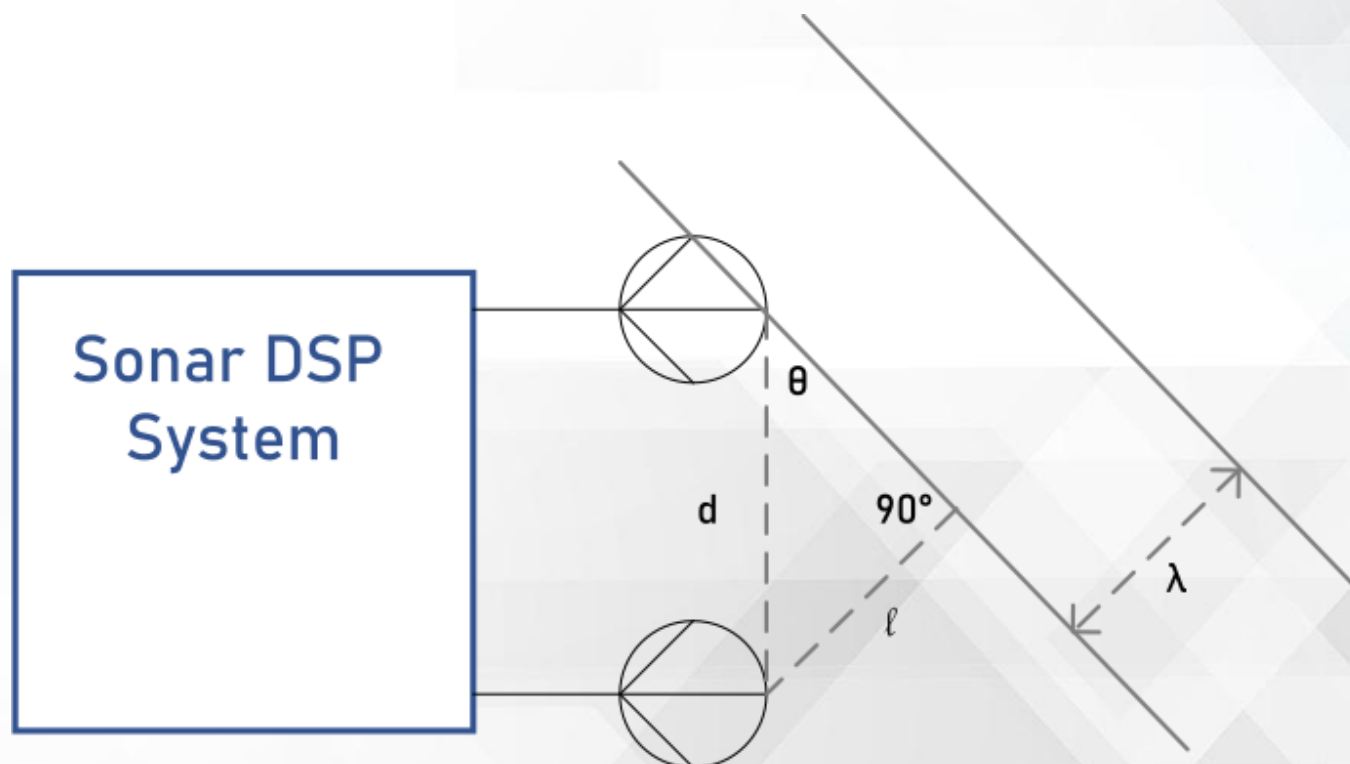


5 – Quartz Material
2 and 4 – Inner and Outer Electrodes, resp.

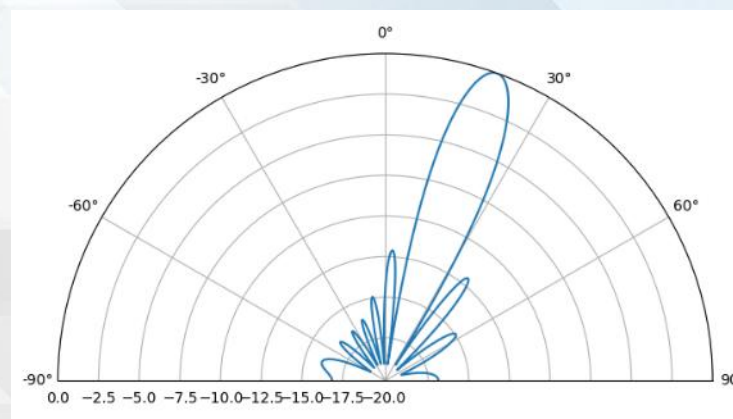


1. B. Bensaude-Vincent and F. Duck, Paul Langevin: Physicist and Social Activist 2025.
2. Naval Sonar, 1953, Available: <https://maritime.org/doc/sonar/>.
3. R. Nightingale et al., "Ultrasound transducer," Radiopaedia.org, 2026, doi: 10.53347/rID-14565.

Beam Patterns and Beamforming



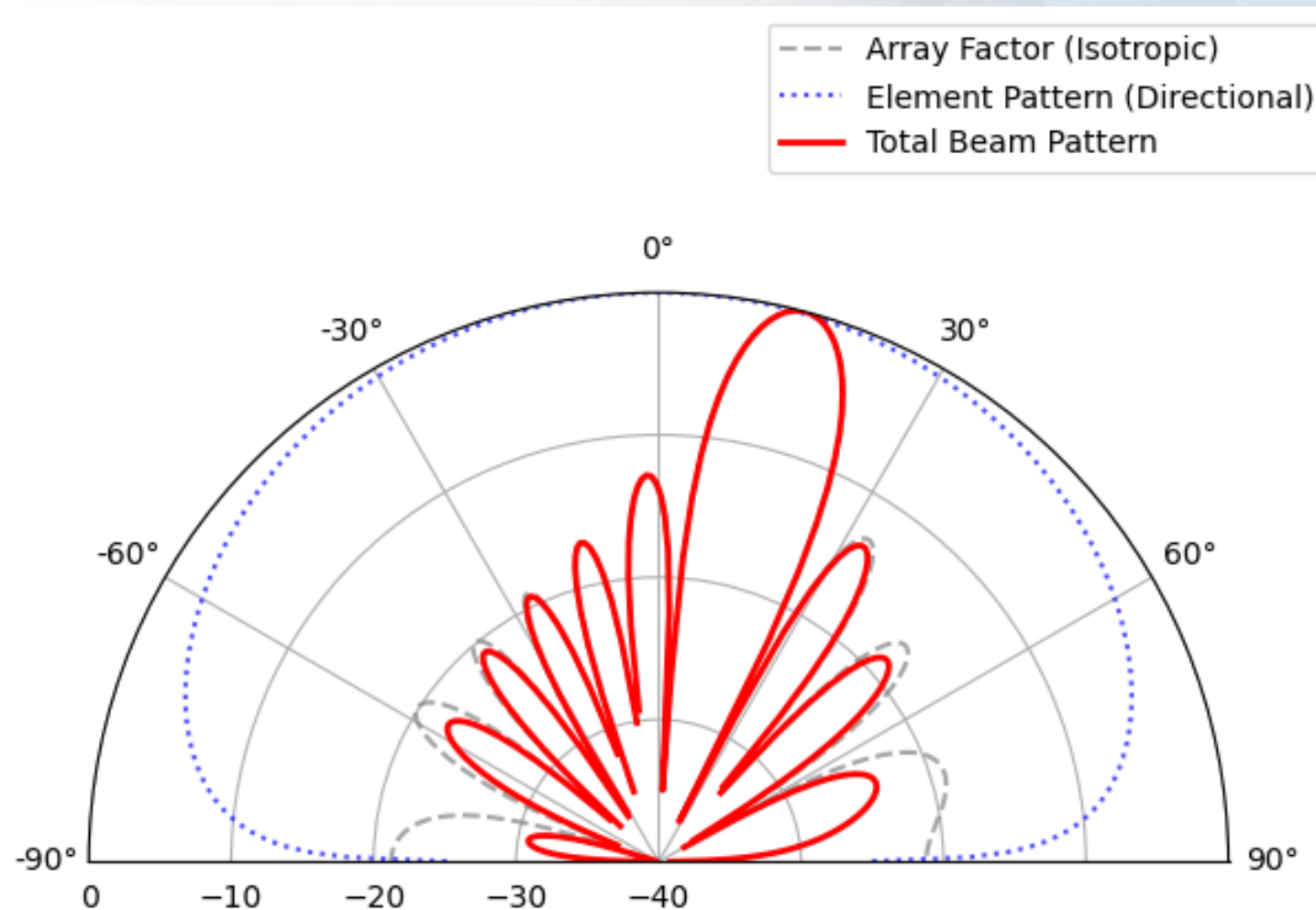
6 Receivers



10 Receivers

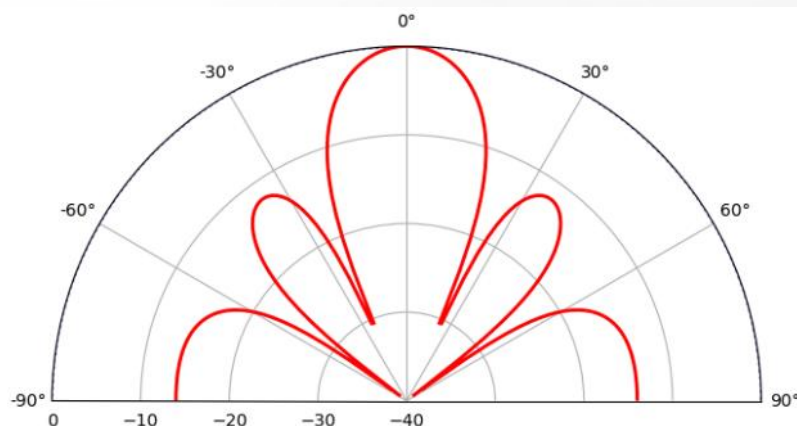
"Beamforming & DOA,"
in *PySDR: A Guide to SDR and DSP using Python*, 2026.

Beam Patterns and Beamforming

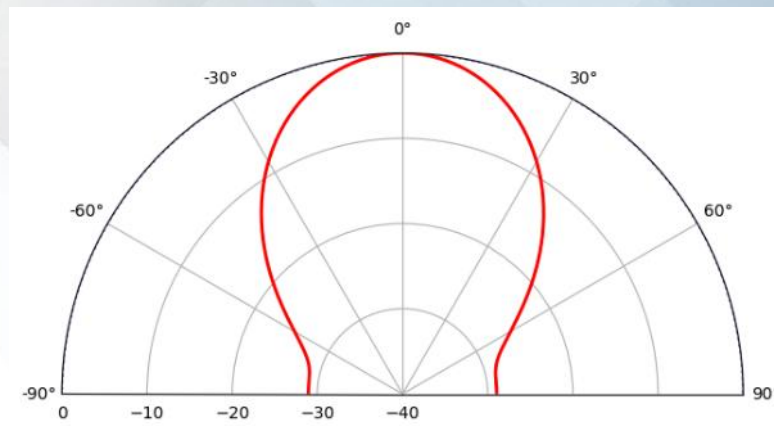


Beamforming – Windowing Functions

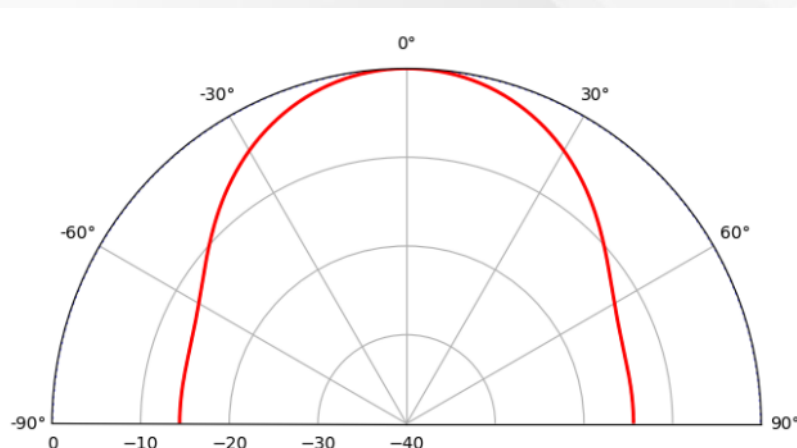
5-Receiver Array Beam Pattern



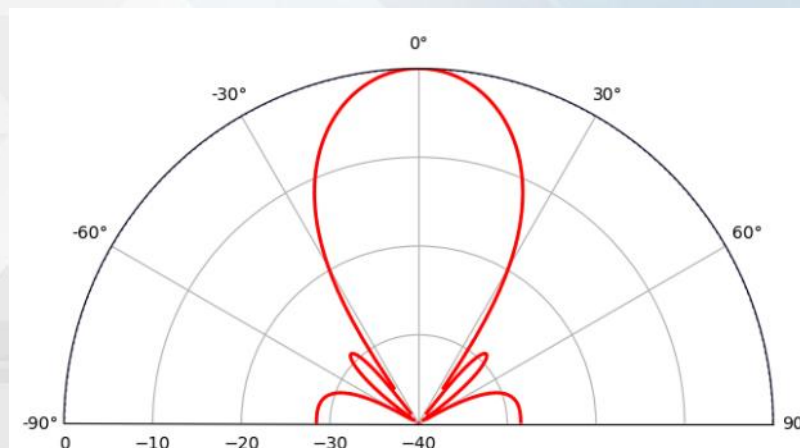
No Window



Hamming Window



Blackman Window

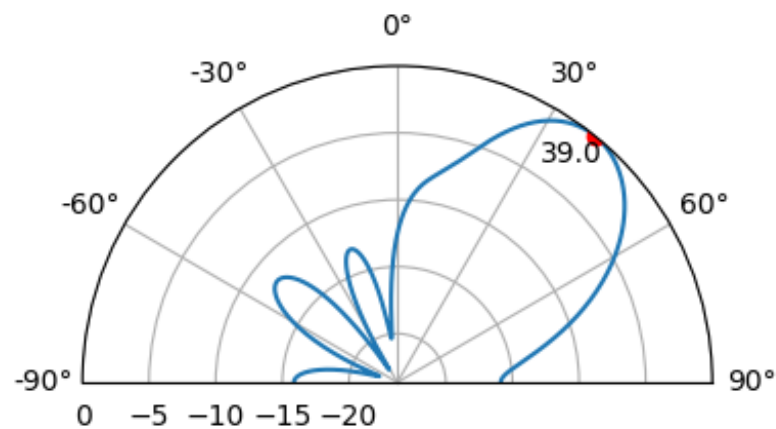


Taylor Window

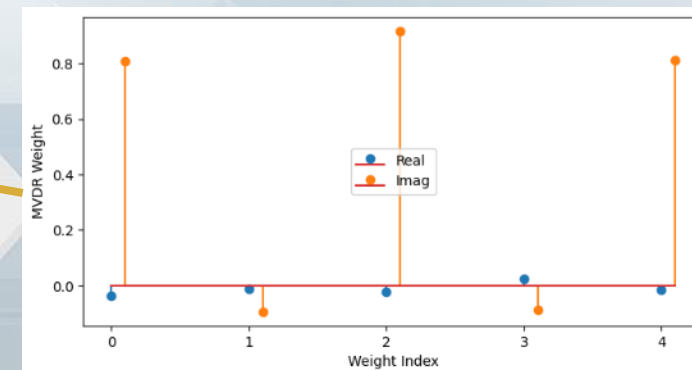
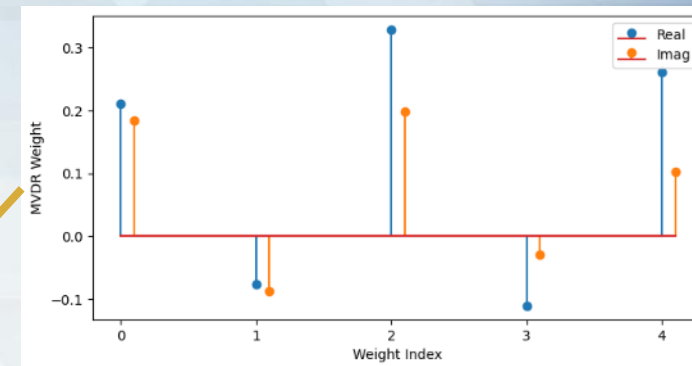
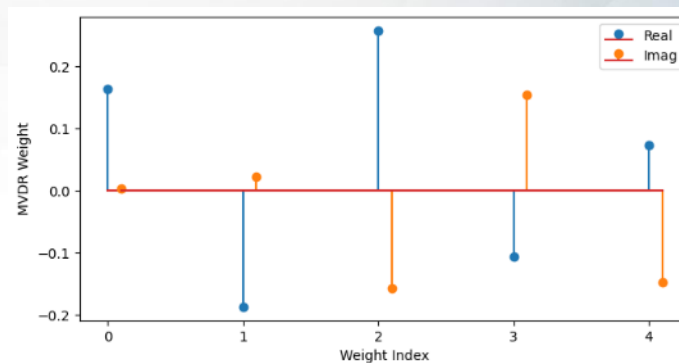
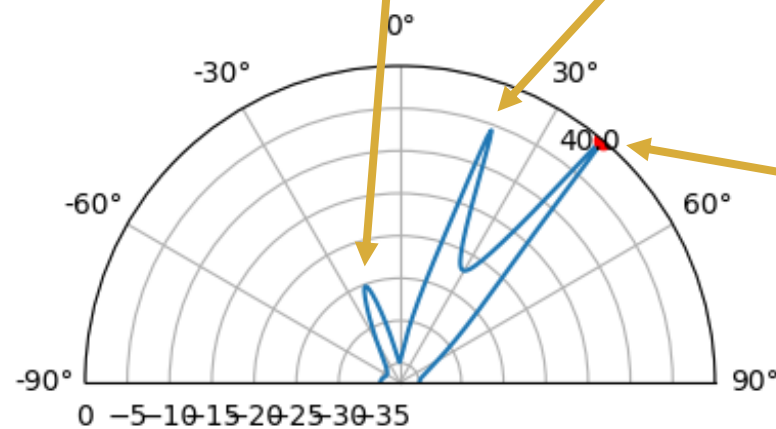
Beamforming – DAS vs. MVDR

(All 1's for weights)

DAS Beamforming



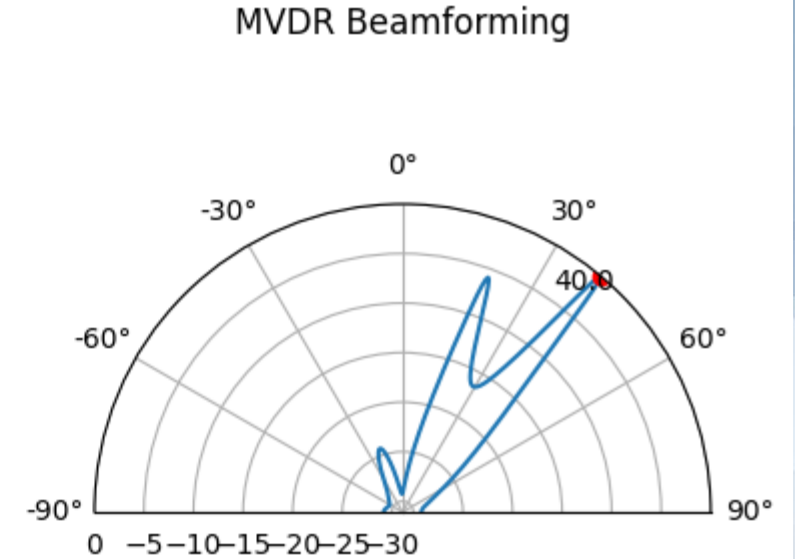
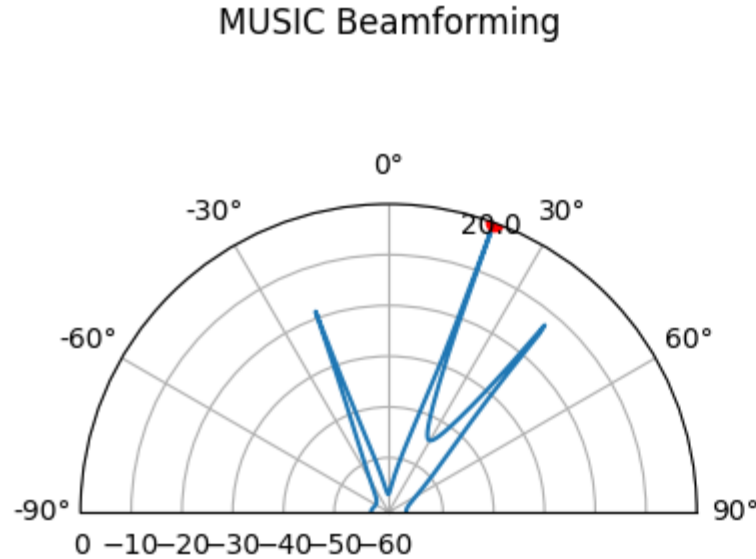
MVDR Beamforming



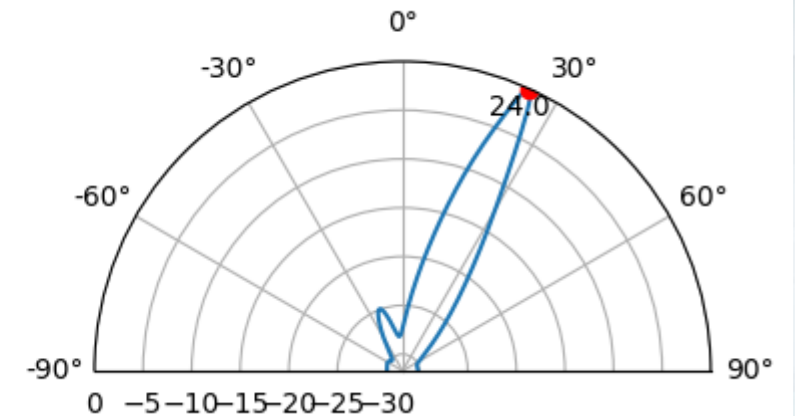
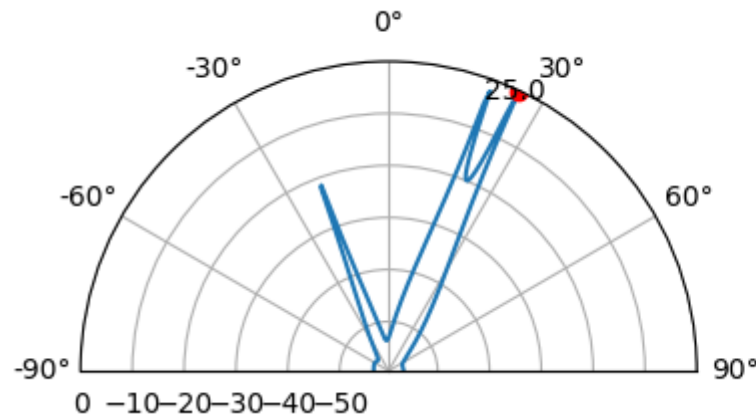
"Beamforming & DOA," in *PySDR: A Guide to SDR and DSP using Python*, 2026.

Beamforming – MUSIC vs. MVDR

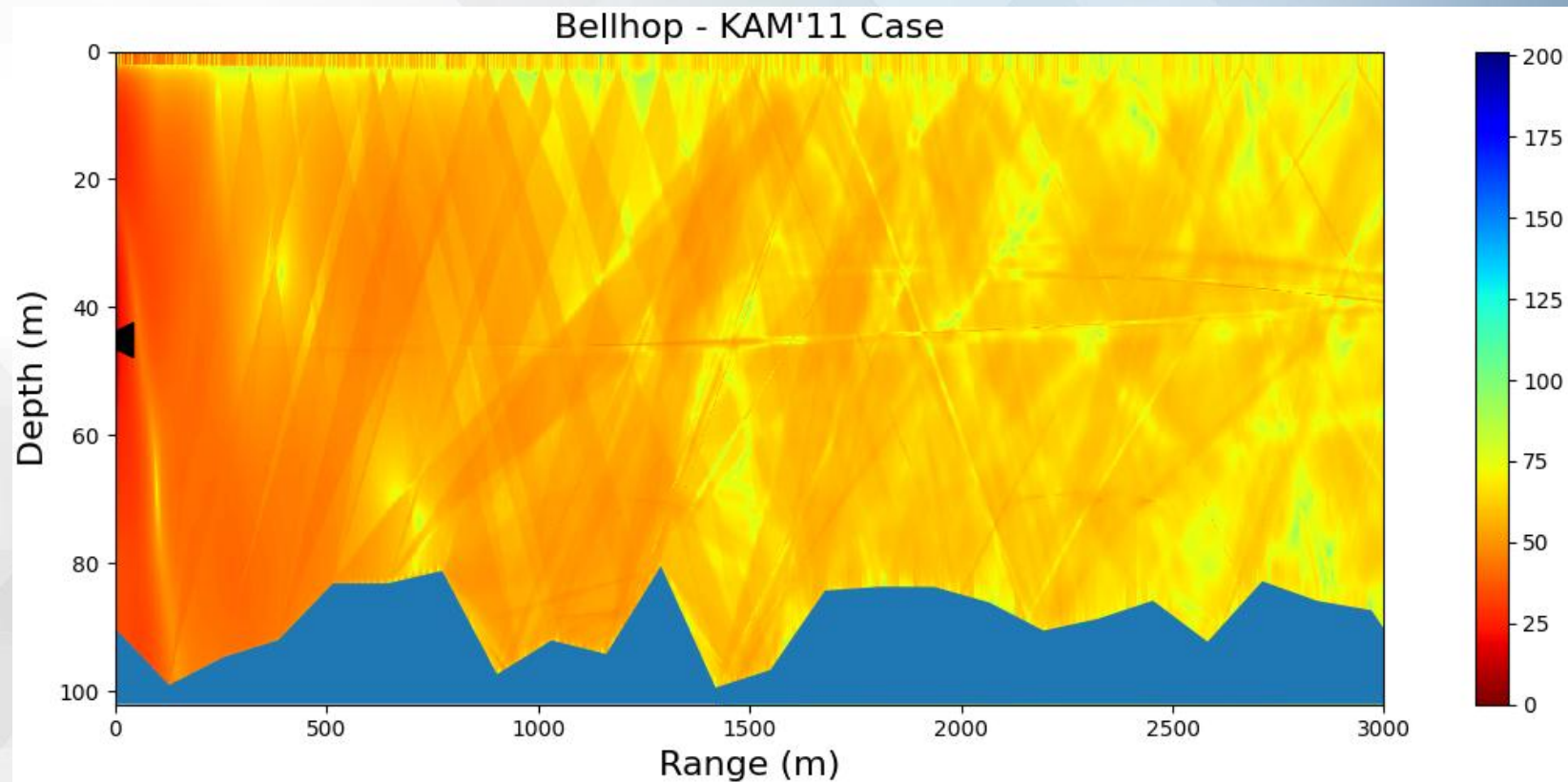
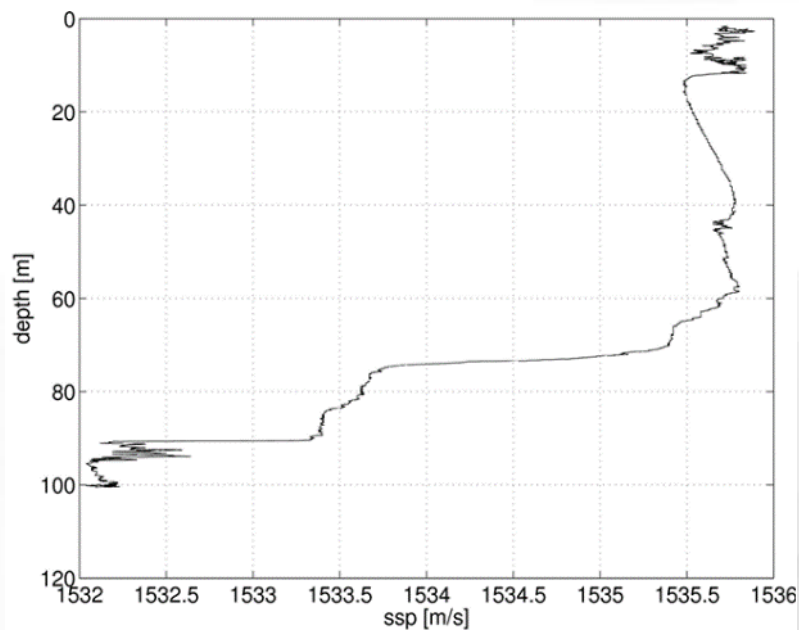
Signals at -20° , 20° , 40°



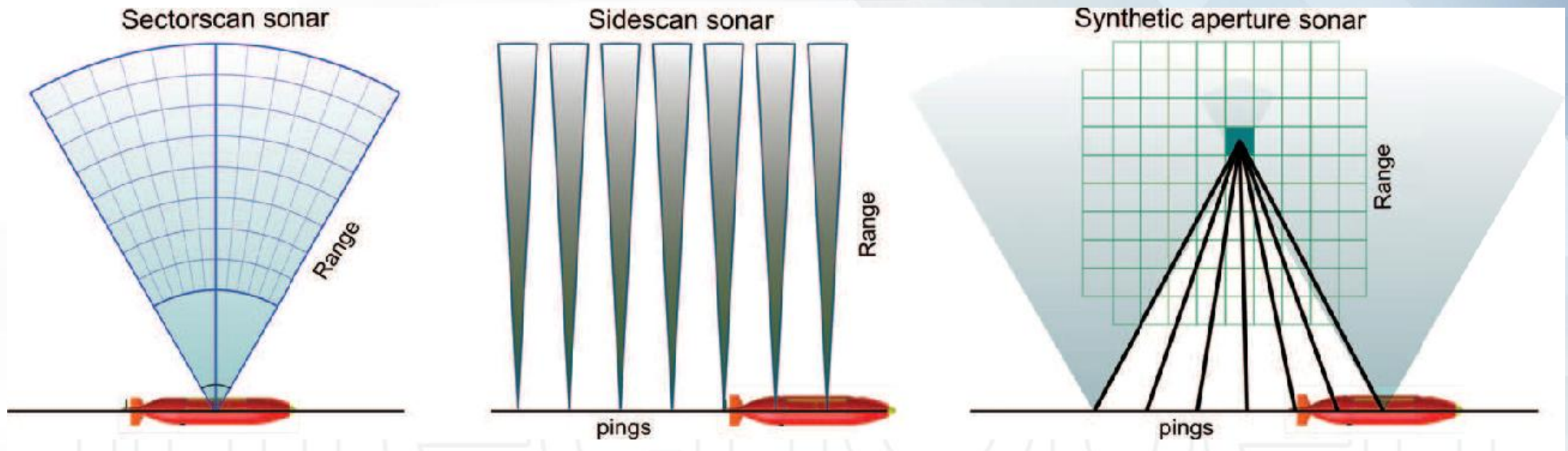
Signals at -20° , 20° , 25°



Beamforming – Interactive Exercise

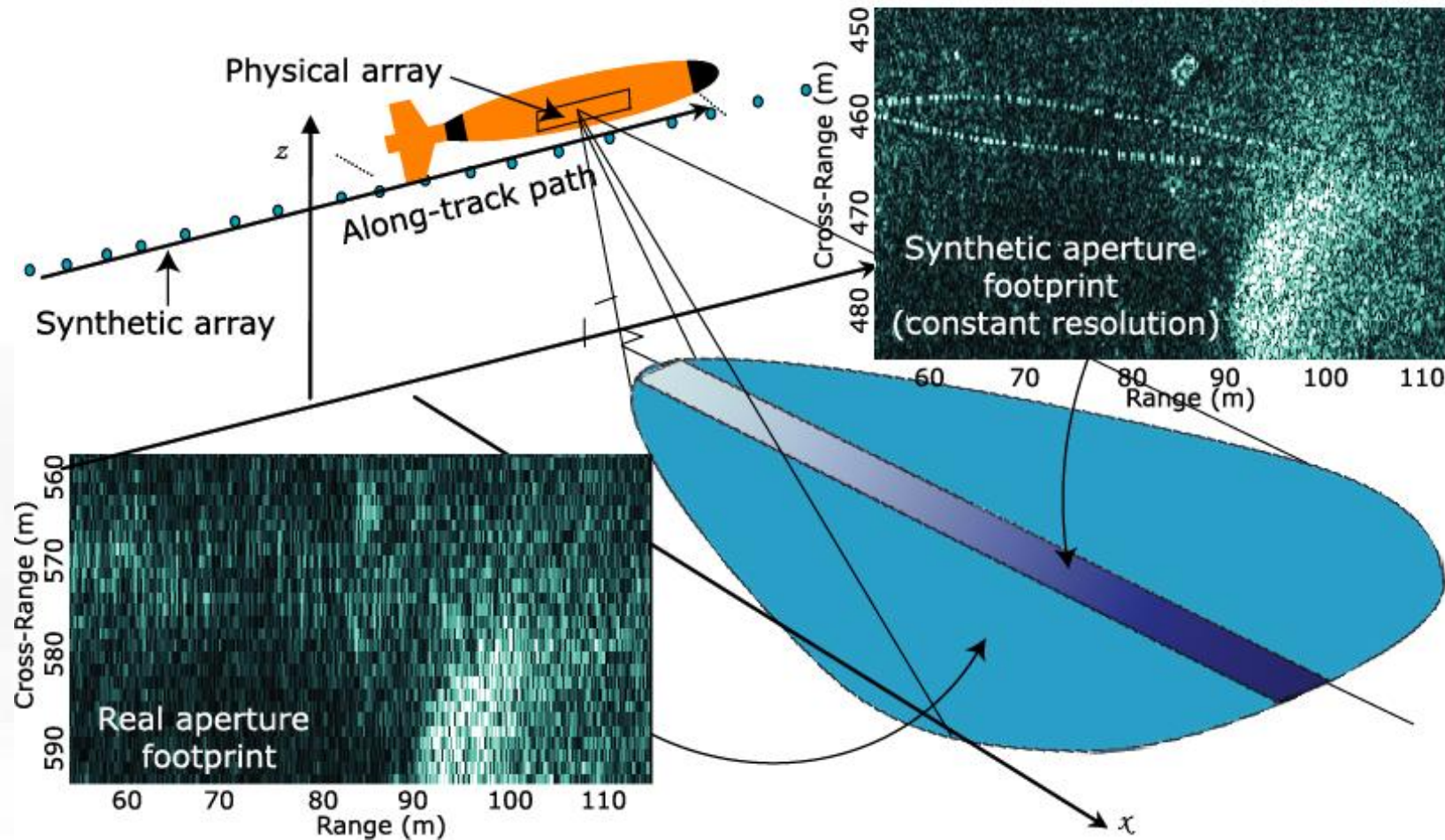


Sidescan Sonar and SAS Imagery



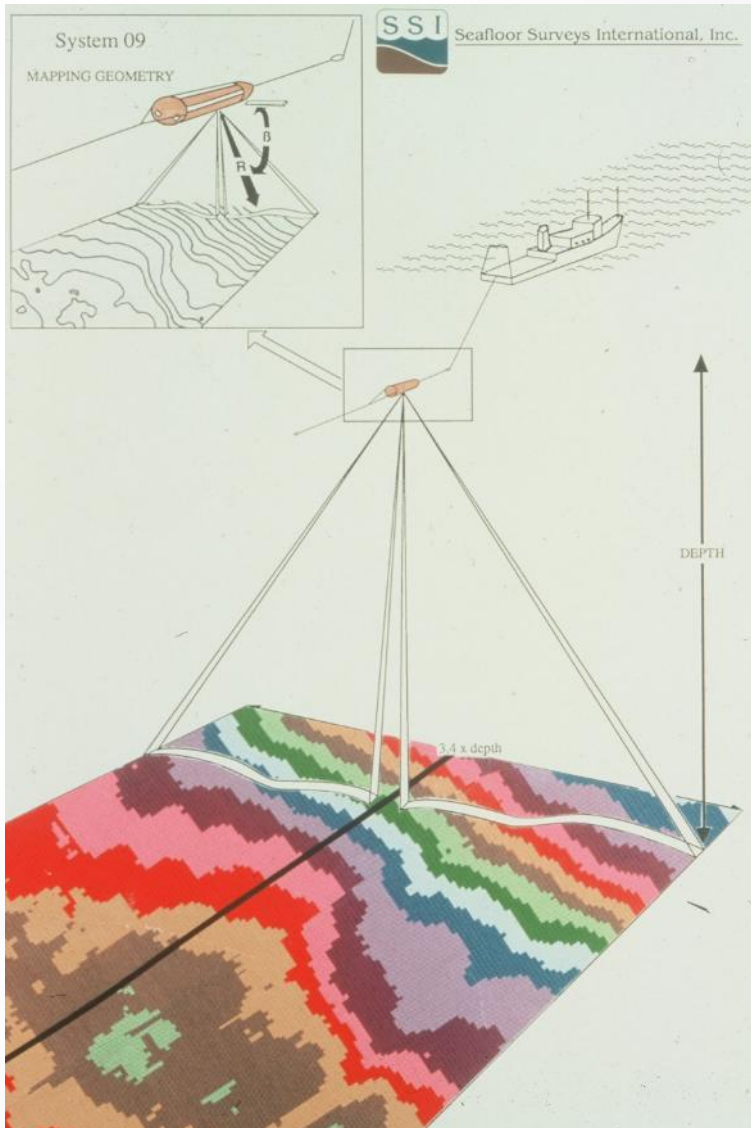
R. E. Hansen, "Introduction to synthetic aperture sonar," in *Sonar Systems*.

Sidescan Sonar and SAS Imagery



D. D. Sternlicht and J. F. Pesaturo, "Synthetic aperture sonar: Frontiers in underwater imaging" (2004)

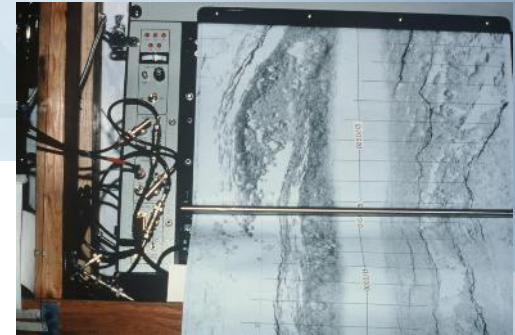
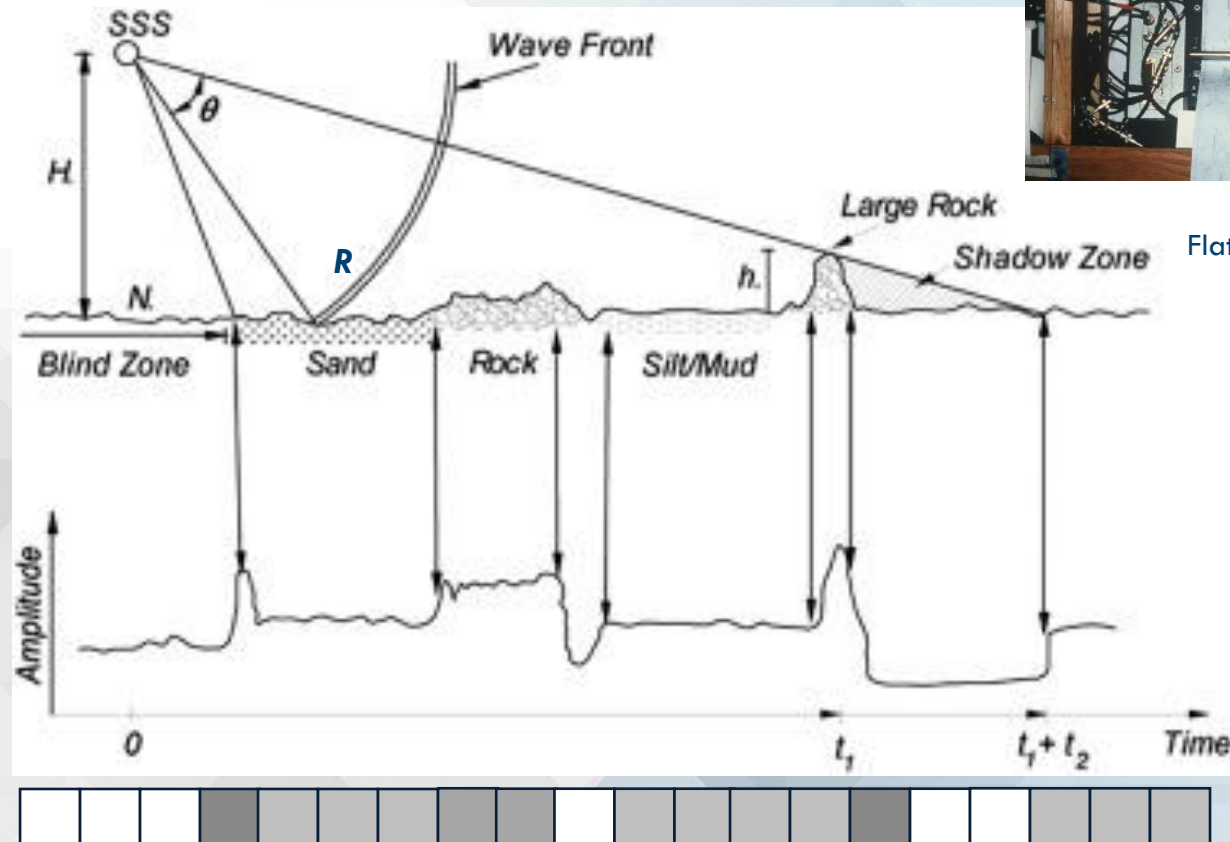
Sidescan Sonar



Seafloor imagery is constructed ping by ping, line by line, as the sonar moves forward. It can be displayed or printed out over time, in what is termed a “waterfall plot.”

$$\text{Range (R)} = \frac{ct}{2}, \text{ where } c = \text{sound speed}$$

$$t = \text{round trip time of return}$$



Flat bottom assumption

Amplitude scan
or A-Scan

This time series of amplitudes represents one sonar ping, and one line of seafloor imagery

Sonar Eqn

Beamforming

Sidescan/SAS

Sidescan Sonar Range (Across-Track) Resolution (SHADOWGRAPH Example)

Sound Speed (c) = acoustic wavelength (λ) X Frequency (f): $c = \lambda f$

Frequency (f): say 1.5 MHz (cycles/second): **acoustic wavelength (λ) = 1 mm**

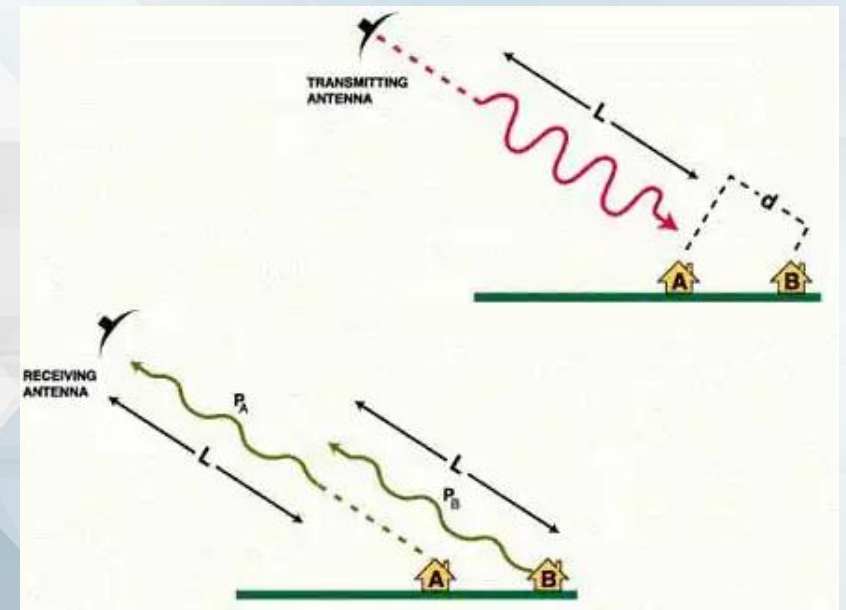
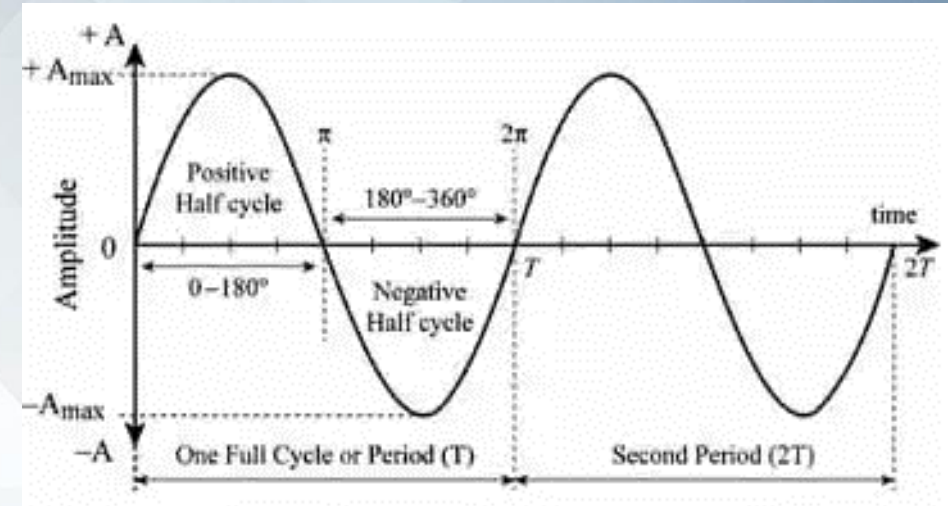
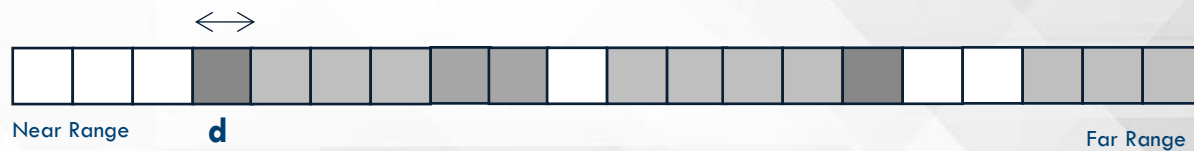
Pulse Length (τ): 0.05 msec in time; or **7.5 cm in length (L) = $c\tau$,**

So about **75 cycles in a Continuous Waveform (CW) pulse**, and a

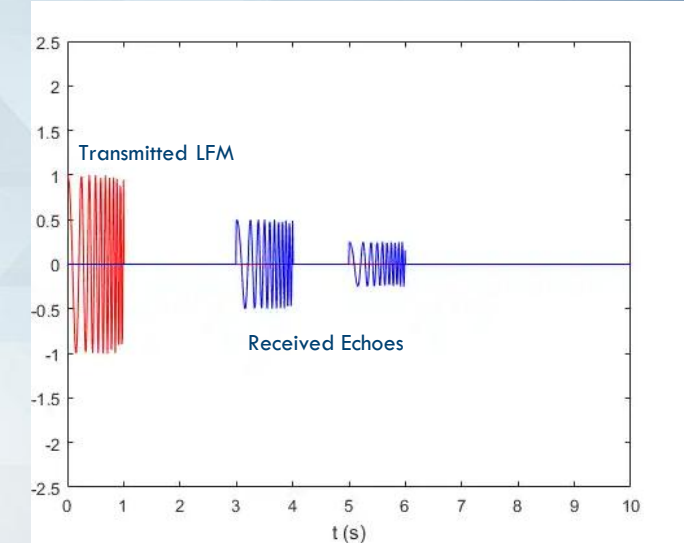
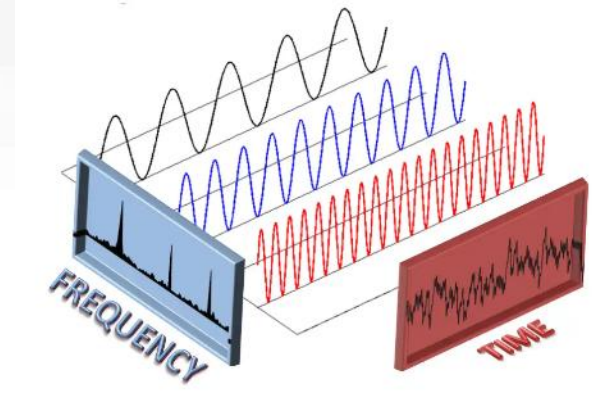
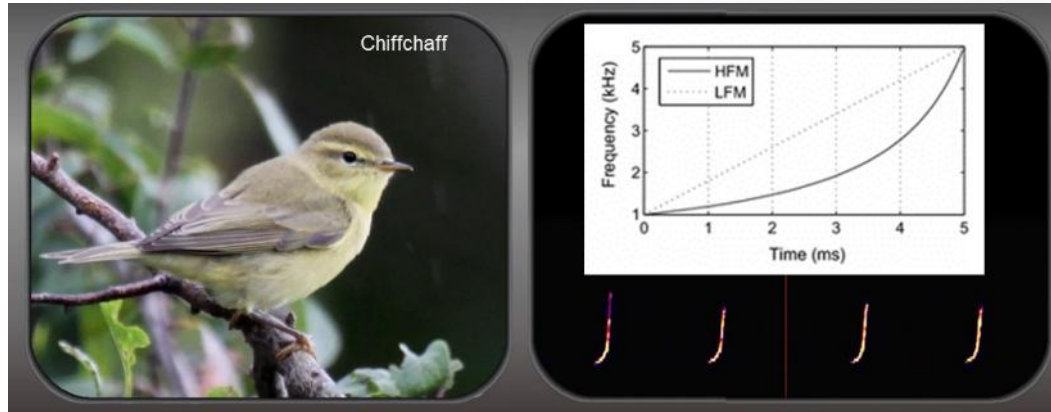
Theoretical range resolution ($d = L/2 = c\tau/2$) of 3.75 cm (~1.5 inch)

i.e. range differential from which you can distinguish echoes from two objects

(i.e. finest pixel length for your sidescan seafloor grid)



FM Techniques to Improve Range Resolution and SNR – “Chirp Sonar”/“Matched Filter”/“Pulse Compression”



Pulse Compression via Cross-correlation

Uncompressed
Received Echo

Cross-correlation

Compressed
Pulse

Transmitted
(Reference)
Signal

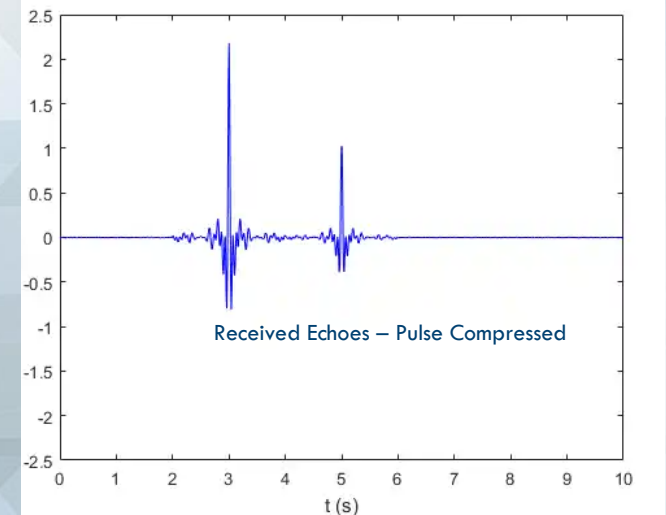
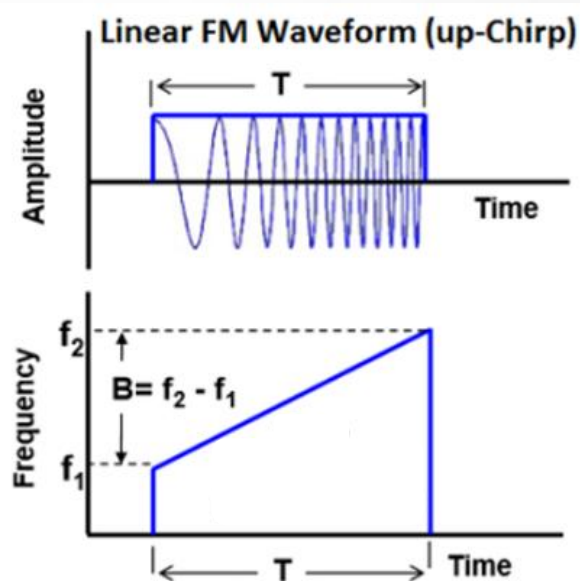
There are fast ways to do this in the Frequency Domain using the *Fourier Transform*!

$$\text{Range resolution} = \text{Speed of Sound} / (2 \times \text{Bandwidth})$$

$$c / 2 B$$

Compare this to CW pulsed system resolution of $c \tau / 2$

For a given range resolution, the pulse can be much longer
(A 20 kHz Bandwidth LFM signal would correspond to the SHADOWGRAPH Example)



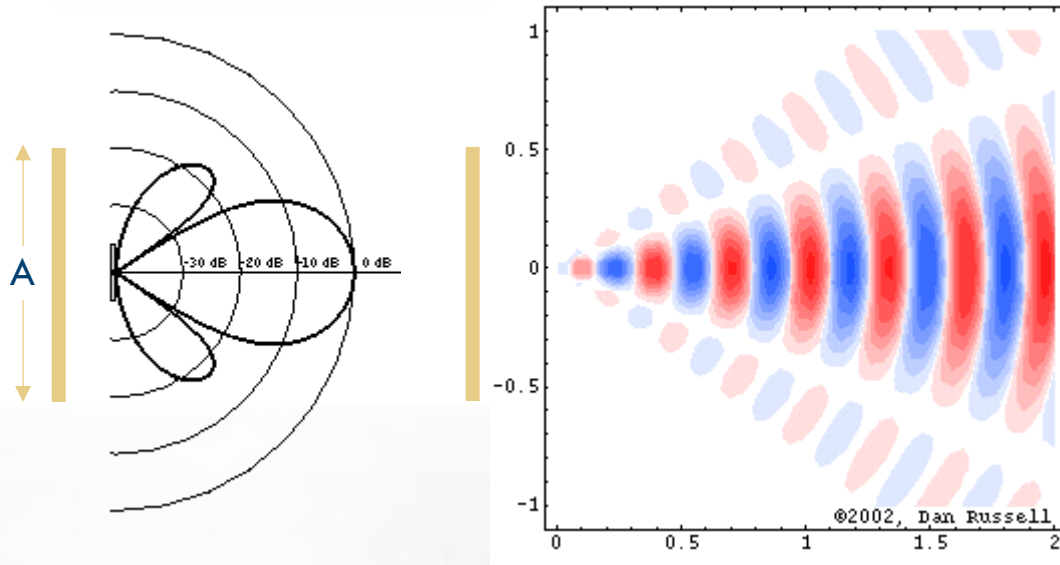
Sonar Eqn

Beamforming

Sidescan/SAS

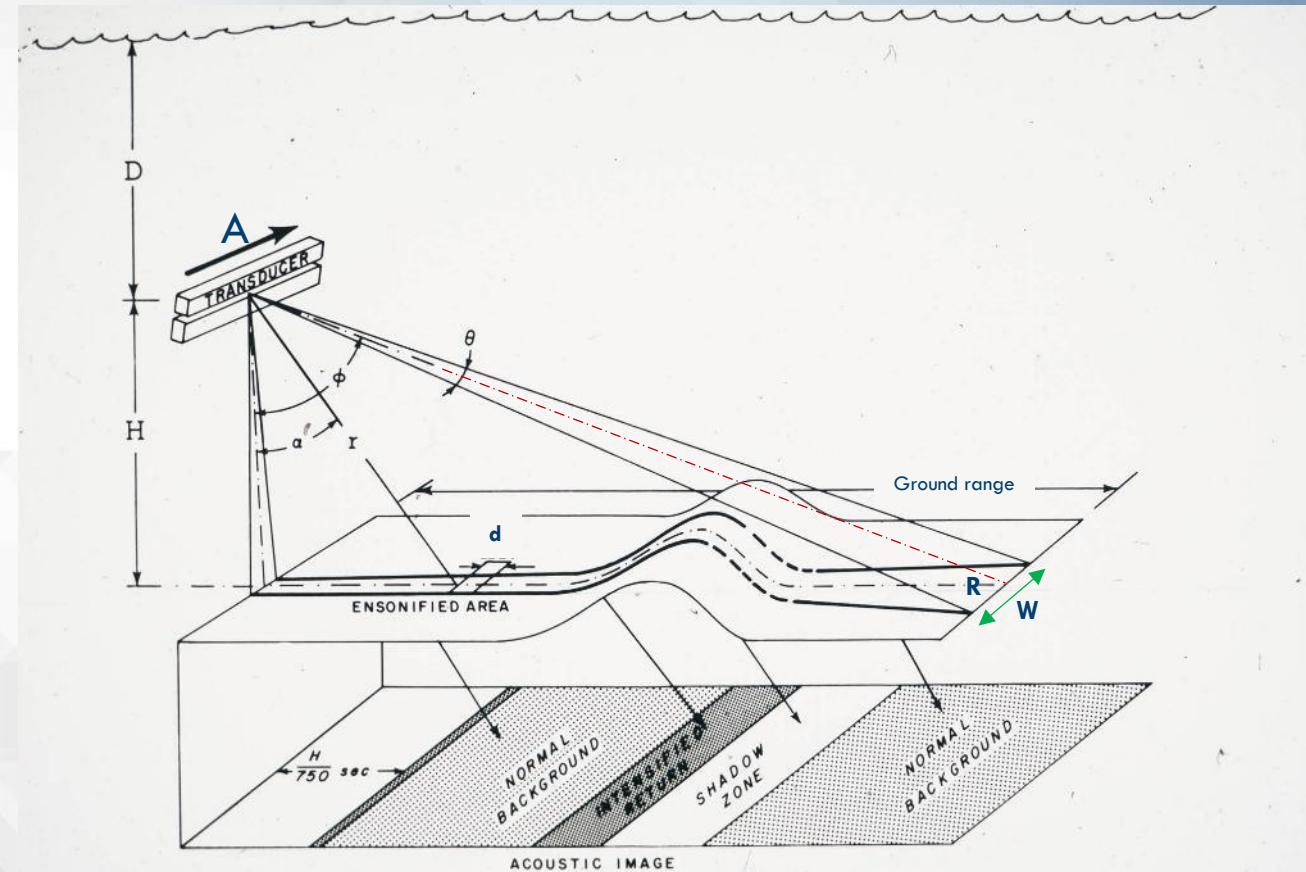
Side Scan Sonar Azimuth (Along-Track) Resolution

Aperture (A), Frequency (f), and Range (R) Determine Along-Track Resolution



Beam Width (θ) $\sim$ Acoustic Wavelength / Aperture
 λ / A

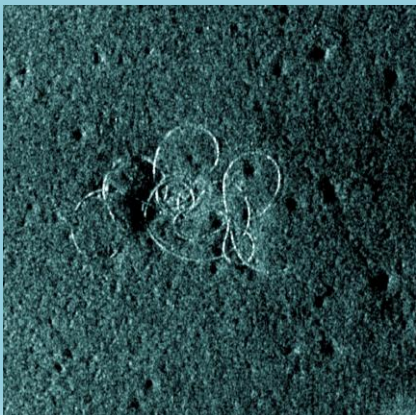
Theoretical Azimuth resolution (W) $\sim R \theta \sim R \lambda / A$
 (i.e. effective pixel width increases with range)



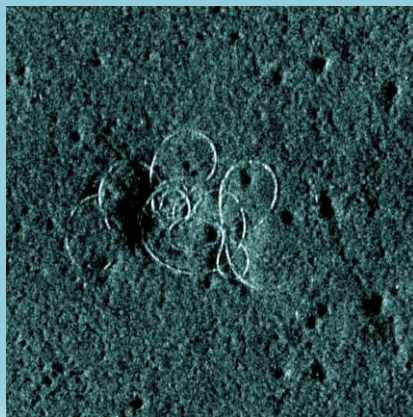
Finer resolution with shorter range, higher frequency, longer aperture

SAS Imagery: Comparison w/Sidescan

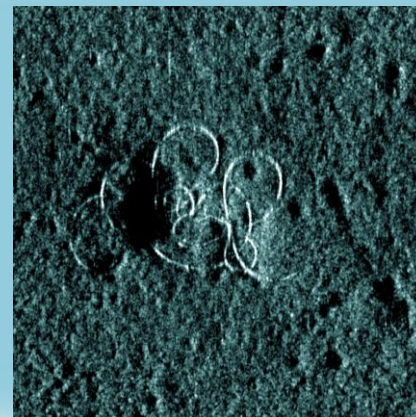
Coil of Wire off Santa Barbara. Depth: 60m



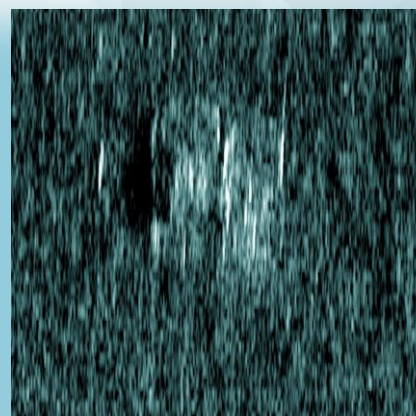
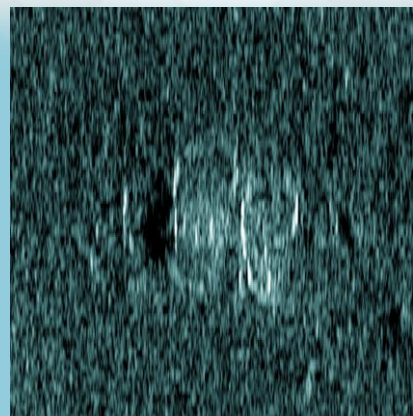
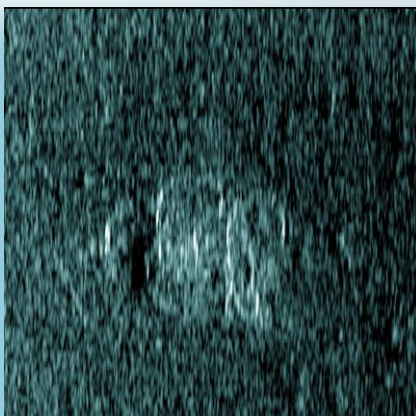
Range: 68m



Range: 81m



Range: 120m



Brute Force: TD-Beamforming Mathematics

1 – Point Range

$$R0 = \sqrt{(\mathbf{u}_0 - \mathbf{P})^T (\mathbf{u}_0 - \mathbf{P})}$$

$$R1 = \sqrt{(\mathbf{u}_1 - \mathbf{P})^T (\mathbf{u}_1 - \mathbf{P})}$$

2 – Time Delay

$$\tau(R0) = 2 \frac{R0}{c} \text{ corresponds to... } x_0(t) \rightarrow X_0(\Omega)$$

$$\tau(R1) = 2 \frac{R1}{c} \text{ corresponds to... } x_1(t) \rightarrow X_1(\Omega)$$

3 – Invert Delay

$$X_{s0}(\Omega, \tau_0) = \sum_{f=-\infty}^{\infty} X_0(\Omega_f) \exp\{2\pi f j \tau(R0)\}$$

$$X_{s1}(\Omega, \tau_1) = \sum_{f=-\infty}^{\infty} X_1(\Omega_f) \exp\{2\pi f j \tau(R1)\}$$

4 – Summation

$$TD(\mathbf{P}) = \sum_{u_k} E[X_{sk}(\Omega, \tau_k)]$$

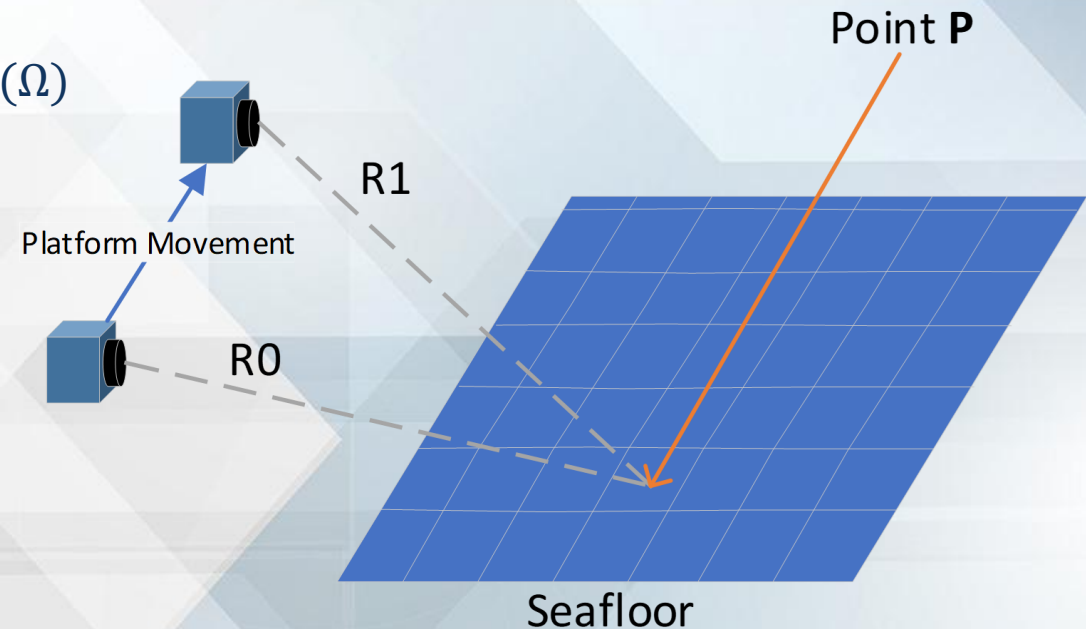
5 – Magnitude

$$Y = 10 \log(|TD|^2)$$

$$\begin{aligned} \mathbf{P} &= (x_P, y_P, z_P) \\ \mathbf{u}_0 &= (a_0, y_0, z_0) \\ \mathbf{u}_1 &= (a_1, y_1, z_1) \end{aligned}$$

Ideal conditions:

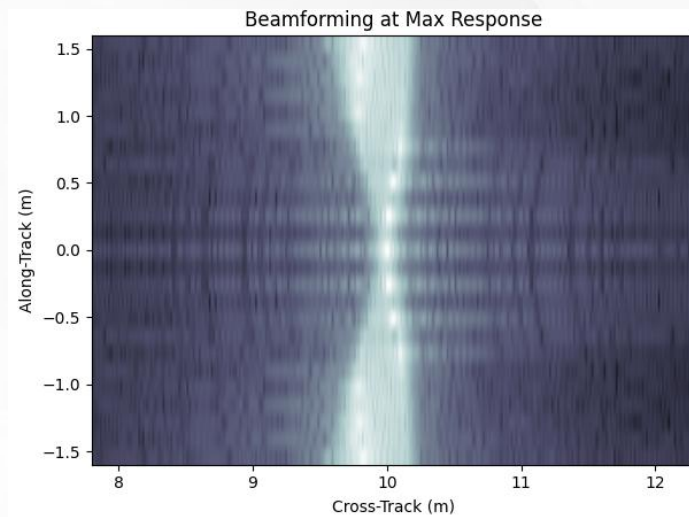
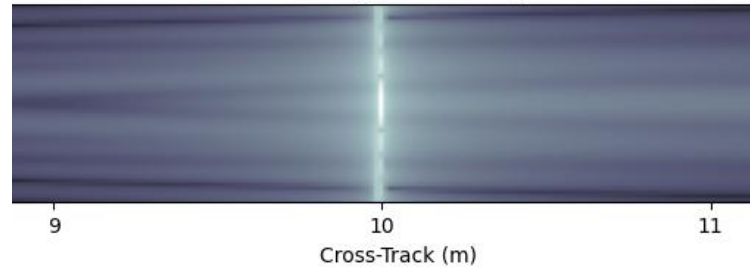
$$a_0 = a_1, z_0 = z_1$$



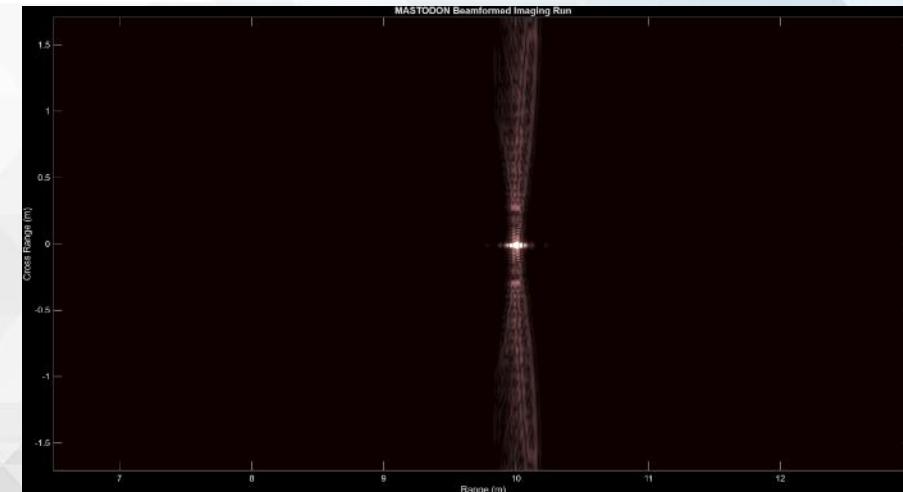
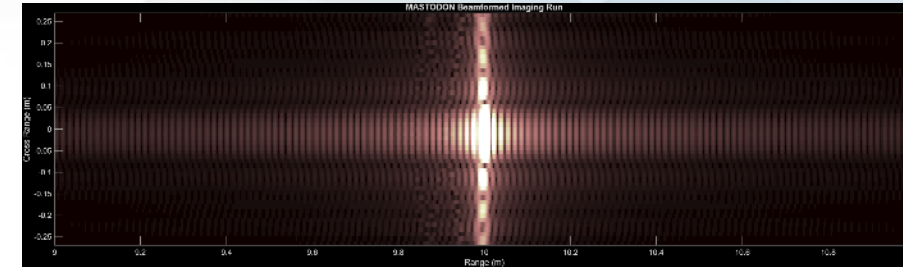
SAS Imagery: TD-Backprojection and Omega-K Results

TD-Backprojection

Beamforming at Max Response



Omega-K



Data created using example code in <https://github.com/Sonar-Sim/MASTODON>.

References

1. L. Hatch, "Live for listening: Underwater sound in national marine sanctuaries," National Marine Sanctuaries, National Oceanic and Atmospheric Administration (NOAA), Apr. 2022. [Online]. Available: <https://sanctuaries.noaa.gov/news/apr22/new-sanctsound-portal.html>.
2. "What is a hydrophone?," National Ocean Service, National Oceanic and Atmospheric Administration (NOAA). Accessed: Sep. 1, 2026. [Online]. Available: <https://oceanservice.noaa.gov/facts/hydrophone.html>.
3. M. A. Richards, *Fundamentals of Radar Signal Processing*, 2nd ed., New York, NY, USA: McGraw-Hill Education, 2014.
4. D. P. Williams and A. J. Hunter, "Multi-look processing of high-resolution SAS data for improved target detection performance," in Proc. IEEE Int. Conf. Image Process. (ICIP), Quebec City, QC, Canada, Sep. 2015, pp. 153–157.
5. "Pulse compression," Wikipedia. Accessed: Sep. 1, 2026. [Online]. Available: https://en.wikipedia.org/wiki/Pulse_compression.
6. B. Bensaude-Vincent and F. Duck, *Paul Langevin: Physicist and Social Activist* (Springer Biographies), Cham, Switzerland: Springer, 2025, doi: 10.1007/978-3-031-95260-9.
7. Naval Sonar, NAVPERS 10884, Bureau of Navy Personnel, U.S. Navy, Washington, DC, USA: U.S. Government Printing Office, 1953, ch. 4, 14, pp. 69–101, 251–268. [Online]. Available: <https://maritime.org/doc/sonar/>.
8. R. Nightingale et al., "Ultrasound transducer," Radiopaedia.org, 2026, doi: 10.53347/rID-14565.
9. D. N. Saikia, "The power of pulse compression," Wireless Pi, Jun. 22, 2024. [Online]. Available: <https://wirelesspi.com/the-power-of-pulse-compression/>.
10. "Science of sound," Discovery of Sound in the Sea (DOSITS), Univ. of Rhode Island. Accessed: Sep. 1, 2026. [Online]. Available: <https://dosits.org/science>.
11. GEBCO Bathymetric Compilation Group, "The GEBCO_2024 Grid—A continuous terrain model for oceans and land at 15 arc-second intervals," NERC EDS British Oceanographic Data Centre NOC, 2024, doi: 10.5285/1c44ce99-0a0d-5f40-e063-7086abc0ea0f.
12. HYCOM Consortium, "Global Ocean Forecasting System (GOFS) 3.1: 41-layer HYCOM + NCODA global 1/12° analysis," Center for Ocean-Atmospheric Prediction Studies (COAPS), Tallahassee, FL, USA. Accessed: Sep. 1, 2026. [Online]. Available: <https://www.hycom.org/dataserver/gofs-3pt1/analysis>.
13. M. B. Porter, "Acoustics Toolbox," Ocean Acoustics Library, Heat, Light, and Sound Research, Inc., La Jolla, CA, USA, Dec. 2025. [Online]. Available: <http://oalib.hlsresearch.com/AcousticsToolbox>.
14. R. J. Urick, *Principles of Underwater Sound*, 3rd ed., New York, NY, USA: McGraw-Hill, 1983.
15. M. Lichtman, "Beamforming & DOA," in *PySDR: A Guide to SDR and DSP using Python*, 2026. [Online]. Available: <https://pysdr.org/content/doa.html>.
16. R. E. Hansen, "Introduction to synthetic aperture sonar," in *Sonar Systems*, N. Z. Kolev, Ed., Rijeka, Croatia: InTech, 2011, ch. 1, pp. 3–28, doi: 10.5772/23122.
17. D. D. Sternlicht and J. F. Pesaturo, "Synthetic aperture sonar: Frontiers in underwater imaging," *Sea Technol.*, vol. 45, no. 11, pp. 27–32, Nov. 2004.
18. D. Woods, "An overview of the MASTODON sonar simulator," *J. Acoust. Soc. Am.*, vol. 154, no. 4, suppl., p. A211, Oct. 2023, doi: 10.1121/10.0023310.